BANK OF THAILAND

MONETARY POLICY REPORT

SEPTEMBER 2017

75th

Bank of Thailand

(1942-2017)

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
September 2017

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Apichai Boontherawara	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2017. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

Content

Executive Summary 1

1. The Global Economy 4

Advanced economies

Chinese and Asian economies

Forecast assumptions on trading partners' economic growth

Global financial markets

Commodity prices

2. The Thai Economy 9

2.1 Recent developments 9

Overall economy

Labor market

Inflation

Financial conditions

Exchange rates

Financial stability

2.2 Outlook for the Thai economy 18

Key forecast assumptions

Growth forecast and outlook

Inflation forecast and outlook

Risks to growth and inflation forecasts

BOX: *Distribution of growth dividends: evidence from the labor market*

BOX: *Household debt deleveraging and implications for the economy*

3. Monetary Policy 30

Monetary Policy Committee's decisions in the previous quarter

4. Appendix 34

4.1 Table 34

Dashboard of indicators for the Thai economy

Dashboard of indicators for financial stability

Probability distribution of growth and inflation forecast

4.2 Chart Pack 38

Economic assessment

Financial stability assessment

Executive Summary

1. Global Economy

The global economy was projected to continue expanding. Advanced economies continued to gain further traction driven mainly by expansion in consumption and stronger labor markets. The U.S. economic growth outlook improved further on the back of private consumption given strong labor market conditions and improved household financial positions and consumer confidence. Euro area economies recorded stronger expansion, with growth broadening across most member countries thanks to accommodative monetary policy stance and a recovery in consumer confidence and labor market conditions. Japan also continued to expand as private consumption picked up due to an increase in wages, improved consumer confidence, and accommodative monetary policy. The expansion was also driven by growing exports thanks to global trade recovery. Meanwhile, emerging economies in Asia continued to grow on the back of improvements in exports, which were underpinned by global demand and technological cycle of electronic goods. In addition, private consumption continued upward trajectory as consumer confidence gradually improved, although such improvement was not robust given elevated household debt levels. The Chinese economy would likely grow at roughly the same pace but was expected to gradually slow down as the economic structural reforms began to bear results. Overall, **the growth forecasts for Thailand's trading partners were revised up**. However, there remained risks that warranted monitoring such as uncertainties pertaining to U.S. economic and foreign trade policies, economic and financial stabilities concerns in China, and geopolitical risks.

Most central banks maintained their accommodative monetary policy stances, although a few central banks raised their policy rates. As for the Federal Reserve (Fed), the Fed was expected to commence its balance sheet reduction and gradually raise its federal funds rate, as the U.S. economy was approaching full employment and headline inflation was projected to slowly rise. However, central banks of some Asian countries, namely Vietnam, India, and Indonesia, cut their policy rates in order to support further growth.

Crude oil prices remained stable but would slowly rise going forward in tandem with global economic recovery and production cuts by major oil producers. The Committee therefore maintained the projection for Dubai crude oil price at 50.9 and 52.8 U.S. dollars per barrel in 2017 and 2018, respectively.

2. Recent Economic Developments

The Thai economy gained further traction in the second quarter of 2017. The growth was primarily driven by external demand from both exports and tourism with signs of greater positive spillovers to domestic demand. Meanwhile, private consumption continued to grow in line with spending on services and durable goods. Public spending remained an important growth driver. Moreover, **headline inflation increased somewhat in the first two months of the third quarter** due to higher domestic retail oil prices and energy charges (FT), while core inflation was largely unchanged from the previous quarter.

Financial conditions remained accommodative. Most short-term money market rates remained close to the policy rate. However, Thai government bond yields declined across all tenures, particularly short-term bond yields that fell significantly in July following the reduction of the Bank of Thailand's short-term bond issuances and a suspension of treasury bill issuance by the government at the end of the fiscal year. However, yields began to pick up due to market adjustments. Meanwhile, the decline in medium-term and long-term yields was influenced by external factors including a slower-than-expected rise in the federal funds rate and conflicts in the Korean Peninsula. With regard to financing costs, the new loan rate remained stable at a low level after having declined in the previous period. Private credit expanded due to an increase in loans extended to households and large corporates in export- and tourism-related businesses. In addition, business financing through equity market continued to expand, although funding through the bond market contracted somewhat as matured debts of some telecommunication and energy companies were not replaced by new debt issuances. With regard to exchange rates, the baht appreciated against the U.S. dollar from the previous quarter. This was attributable to the weakening of the U.S. dollar and other domestically driven factors such as higher-than-expected current account surplus and sizable foreign direct investment flows into Thailand in this quarter. Nevertheless, the baht's movement relative to those of trading partners was largely unchanged.

Financial stability remained sound overall but there remained pockets of risks that might result in the buildup of vulnerabilities in the financial system going forward. External stability continued to be strong while financial positions of large corporates and financial institutions remained sound. Financial institutions maintained high levels of capital buffers and loan loss provisions to cushion themselves against risks. Nevertheless, there remained pockets of risks that could potentially lead to the buildup of vulnerabilities in the financial system going forward. These included (1) debt serviceability of both households and businesses that remained fragile, (2) search-for-yield behavior, and (3) oversupply of condominium units in Bangkok and vicinity areas as well as large-scale mixed-use real estate developments.

3. Economic Outlook

Looking ahead, the Thai economy was projected to expand further, recording 3.8 percent in 2017 and 2018. The upward revision to the previous forecast was on account of a continued improvement in merchandise exports and tourism. In addition, private spending gradually rose and was more broad-based while fiscal impulse remained.

The expansion in merchandise exports was more broad-based across product categories and export destinations, with improvements seen across exporting firms of all sizes. The value of merchandise exports in 2017 was projected to grow by 8 percent in tandem with global economic recovery. Exports of electronics would likely expand in line with the technological cycle, while several products also benefited from the relocation of production bases to Thailand. Moreover, export prices were also projected to trend up, especially commodities, in line with crude oil prices. Nevertheless, the exports expansion would likely raise demand for raw materials and intermediate goods, prompting this year's import value to rise further.

Exports of services continued expanding in tandem with stronger performance of the tourism sector. The projection for the number of foreign tourists in 2017 was revised up from 34.7 to 35.6 million due to several reasons: (1) increasing number of Chinese tourists—both group and independent tourists—thanks to the opening of new direct flight routes from China to major tourist destinations in Thailand, (2) a rising number of ASEAN tourists that was in line with economic recovery, and (3) the reduction and exemption of tourist visa fees. Moreover, global economic recovery helped support a further rise in tourism spending per head.

Private consumption would expand at a gradual pace in the period ahead, supported by improvements in farm income and employment in export-related manufacturing and services sectors as well as government measures such as the social welfare card project and the 9101 Project. However, household purchasing power would remain modest going forward because employment and wages had yet to fully benefit from the export recovery, partly due to economic structural changes and business models that were less reliant on labor, together with elevated level of household debt.

Public spending remained an important growth driver. Budget disbursement for both public consumption and investment was well on track despite some unexpected delay in certain projects such as investment projects of state-owned enterprises. Meanwhile, government investment was expected to slow down after prior acceleration, while some government agencies were constrained by limited disbursement efficiency. In addition, the promulgation of the Public Procurement and Supplies Management Act, B.E. 2560 might result in a delayed disbursement, during the initial phase, of some state agencies that would now be governed by the new law, particularly local administrations that had not previously operated under this system.

Private investment continued to recover albeit at a gradual pace. In the near term, investment recovery was observed in various industries, consistent with an expansion in private consumption and exports. However, there remained excess production capacity in some businesses as they were also awaiting for greater clarity in the government's stimulus policies especially those regarding some infrastructure investment projects.

Headline inflation remained low due to supply-side factors but was expected to slowly rise. In recent periods, headline inflation was lower than previously assessed attributed largely to supply-side factors, especially fresh food prices. Moreover, the lower fresh food prices also helped hold down the food component in core inflation at low level. Going forward, inflation was expected to edge up slowly on the back of the gradual increase in demand-pull pressures given the improved growth outlook and higher cost-push pressures from an increase in excise tax, higher wages from the new regulations on immigrant workers, and the minimum wage rise next year. However, structural factors such as e-commerce and globalization trends, which intensified competitions among businesses, coupled with lower production costs driven by technological advancements, would likely cause inflation to rise at a gradual pace overall. **The Committee therefore revised down its forecasts for headline inflation to 0.6 and 1.2 percent in 2017 and 2018, respectively.**

Risks to growth became more balanced in the near term, albeit still tilted to the downside, while risks to inflation were judged to be balanced. Uncertainties surrounding the growth forecasts decreased given the better growth outlook of Thailand's trading partners. However, there were possibilities that growth outturn might be lower than the baseline projection owing to uncertainties pertaining to U.S. foreign trade policy, China's ongoing economic structural reforms, and geopolitical risks, all of which could have adverse impact on Thailand's trading partners. On the domestic front, downside risks stemmed from household purchasing power that was not yet broad-based, the stringent regulations on immigrant workers that might affect economic activities, and a possible slowdown in public spending during an initial phase of Public Procurement Act adoption. On the upside, there were possibilities that the Thai economy would achieve a higher growth than the baseline projection given a stronger-than-expected growth of the U.S. economy due to domestic stimulus measures. Unlike the balance of risks to growth, the risks to inflation were projected to be in balance, with downside risks that inflation might fall below the baseline projection coming from a lower-than-projected economic growth and the impact of structural changes from technological advancements and

intensifying business competitions. However, on the upside, inflation might be higher than baseline projection should regulations on immigrant workers result in a tightening labor market and wages increases.

4. Monetary Policy Deliberation

In the Monetary Policy Committee meetings on August 16 and September 27 2017 the Committee unanimously voted to maintain the policy interest rate at 1.50 percent. In deliberating their policy decision, the Committee assessed economic growth to continue to gain further traction, driven mainly by external sectors with signs of greater positive spillovers across various sectors of the economy. Domestic demand started to improve; however, supporting factors to consumption were not yet robust. Meanwhile, headline inflation was projected to be below the lower bound of the target this year due to supply-side factors, especially the decline in fresh food prices and structural factors that were observed in many countries. However, inflation was projected to slowly rise in tandem with economic growth and domestic demand recovery.

Overall financial conditions remained accommodative and conducive to economic growth. Short-term government bond yields declined mainly as a result of the reduction in short-term bond issuances by both the Bank of Thailand and the Ministry of Finance and did not indicate future monetary policy stance. Meanwhile, the Thai baht appreciated somewhat against the U.S. dollar, but the baht's movement was largely unchanged relative to those of trading partners. Nevertheless, the baht experienced stronger appreciation compared with regional currencies in some periods due to Thailand's external positions and other specific domestic factors. Going forward, the Committee viewed that the baht might experience high volatilities due to uncertainties from the external front. Therefore, the Committee would closely monitor developments in the foreign exchange market.

Financial stability remained sound overall, but there remained pockets of risks that might lead to the buildup of vulnerabilities in the period ahead. These risks included deterioration in debt serviceability of households and SMEs stemming from elevated debt levels and structural problems that weighed on SMEs' adjustments amid a changing environment in business competitions. In addition, even though the situation regarding the rollover of the unrated corporate bonds improved, there were still pockets of risks that warranted monitoring including the debt serviceability of bonds that would mature in the period ahead and maturity mismatch in business financing structures, especially those in the real estate sector. In addition, the prolonged low interest rate environment not only could affect national savings but also could increase the search-for-yield behavior, as reflected in a continued increase in investments in financial assets abroad that were concentrated in some countries as well as the search-for-yield behavior of saving cooperatives. Consequently, these might lead to widespread underpricing of risks. The Committee would thus continue to monitor developments of such risks; furthermore, the Committee viewed that developments of financial market infrastructure and collaboration among regulatory authorities would be crucial in enabling better information access for risk assessment and preventing any exploitation of the regulatory gap. The Committee would stand ready to implement appropriate macroprudential measures in a timely manner.

Monetary policy should remain accommodative to support the continuation of economic growth and spillovers to domestic demand, which would help increase inflationary pressures in the period ahead. The Committee assessed that Thailand's economic growth continued to gain further traction on the back of both external and domestic factors, though there remained risks to the external front that could affect export and tourism growth. Moreover, domestic demand that was not yet sufficiently strong, coupled with certain supply-side and structural factors, led to low inflationary pressures. Meanwhile, financial conditions remained accommodative with sound financial stability. The Committee viewed that the degree of current monetary policy accommodation should be maintained to support further economic growth and spillovers to domestic demand. These would facilitate the increase in inflationary pressures and the return of headline inflation to target, although this might take some time. The Committee would stand ready to utilize an appropriate mix of available policy tools to support economic growth while ensuring financial stability.

1. The Global Economy

Advanced economies continued to gain further traction, driven mainly by expansion in consumption and a stronger labor market.

The U.S. economy continued to expand owing to private consumption thanks to improved economic fundamentals, stronger labor market conditions, and improved household financial positions and consumer confidence. Investment continued to gradually recover from private investment given strong business sentiment. Meanwhile, Hurricanes Harvey and Irma would have a negative impact on the U.S. economy only for a short period of time. Besides, after the Congress approved a suspension of the debt ceiling for a three-month interval from September 8 to December 8 2017, the government shutdown and debt default were less likely to occur. **Euro area economies recorded stronger expansion, with growth broadening across most member countries**. Accommodative financial conditions and improved consumer confidence and labor market conditions would support a consumption growth going forward. **Japan continued to expand** on the back of stronger private consumption growth. The expansion was partly due to base pay raises at the annual wage negotiations between management and unions, robust consumer confidence, and accommodative monetary policy stance. Moreover, export expanded in line with a global trade recovery.

Looking ahead, the expansion of advanced economies would still face risks from (1) uncertainties pertaining to U.S. economic policies on foreign trade, tax policy reform and infrastructure investment, and (2) negotiations on trade and other issues between the U.K. and the European Union after the Brexit.

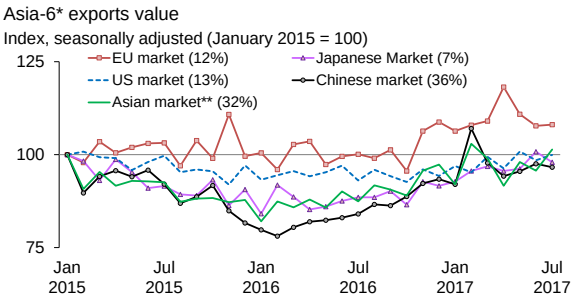
Asian economies exhibited a stronger growth on the back of exports thanks to improvements in global demand and a gradual domestic demand recovery. Meanwhile, the Chinese economy continued to gradually slow down following ongoing economic structural reforms.

China's growth in the second quarter of 2017 remained close to the previous quarter on account of continued expansion in investment and exports. Production and investment in the manufacturing sectors were underpinned by global demand recovery. Meanwhile, public investment growth slowed down somewhat, particularly for state-owned enterprises resulting from ongoing economic structural reforms. In the period ahead, China's growth would continue to expand albeit at a gradual pace due to (1) the on-track progress of economic structural reforms and (2) financial stability measures. In particular, measures on the property sector implemented at the end of last year would gradually slow down investment on the property sector going forward. However, global demand recovery and the implementation of the One Belt One Road project would support production and export growth in the period ahead. Nonetheless, there remained risks to China's financial stability that warranted close monitoring, especially those regarding high corporate debt levels and capital outflows, as these factors could weigh on China's economic growth.

Asian economies (excluding Japan and China) continued to expand thanks to exports especially of electronics that benefited from greater external demand and the upward technology cycle (Chart 1.1). The expansion of exports started to have positive spillovers to investment in machinery and equipment and led to higher employment in manufacturing sector in

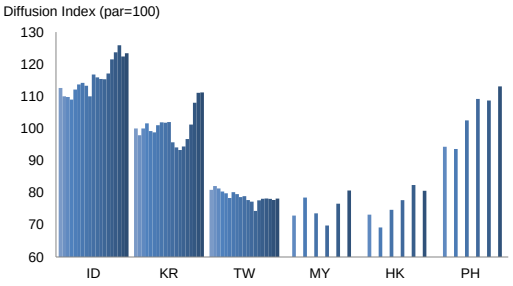
some certain countries. In addition, although private consumption picked up following gradual improvements in consumer confidence (Chart 1.2), it was not yet sufficiently strong owing to elevated household debt. Nevertheless, the Asian economic outlook could be weighed down by (1) China's growth that was expected to slow down and its implications on Asian exports, (2) elevated household debt in many countries that could constrain domestic demand recovery, and (3) risks pertaining to the U.S. foreign trade policies as well as shifts in global trade structure, whereby advanced economies and China could rely more on domestic production and relocate their production sites to closer regions. Consequently, these could undermine the export growth outlook—a main growth driver of the Asian economies in the period ahead.

Chart 1.1 Asian exports continued to increase in the second quarter.



Note: *Asia-6 includes Hong Kong, Taiwan, S. Korea, Malaysia, Singapore, and Thailand.
**Asian market includes Hong Kong, Taiwan, S. Korea, Singapore, Philippines, Indonesia, and Thailand.
Source: CEIC and calculations by Bank of Thailand

Chart 1.2 Asian consumer confidence continued to improve.



Note: Philippines' consumer confidence is scaled from par at 0 to 100
Source: CEIC

The growth prospect for Thailand's trading partners was revised upward, but there remained risks that still warranted monitoring.

Economic growth of Thailand's trading partners would likely to exhibit stronger than expected in the previous *Monetary Policy Report* (June 2017). This was partly attributable to the better-than-expected economic data outturns in the second quarter of 2017, improved private concumption in advanced economies, and continued Asian export growth. The Committee therefore revised up the growth forecast for Thailand's trading partners to 3.6% and 3.4% in 2017 and 2018, respectively (Table 1.1).

The Committee assessed that Thailand's trading partners would still face risks that warranted monitoring as the implementation of U.S. economic stimulus would be less likely and geopolitical risks heightened following tensions in the Korean Peninsula. Nonetheless, some certain risks started to diminish. Such risks included concerns on China's financial stability that were declined after the credit tightening was implemented and the Committee to ensure financial stability was established to support close collaboration among regulators. Another risk was uncertainties surrounding the U.S. foreign trade policies that were reduced. This was because U.S. government was more likely to settle on multilateral economic and legal agreements which might take some time.

Table 1.1 Assumption on trading partners' economic growth

Annual change (%YoY)	Weight (%)	2016*	2017	2018
United States	14.9	1.6	2.1 (2.1)	2.1 (2.1)
Euro area	10.0	1.7	2.1 (1.8)	1.6 (1.5)
Japan	13.6	1.0	1.5 (1.2)	1.1 (1.0)
China	15.7	6.7	6.7 (6.6)	6.3 (6.2)
Asia (excluding Japan and China)**	37.4	3.5	4.2 (3.8)	3.9 (3.8)
Total***	100	3.1	3.6 (3.4)	3.4 (3.3)

Note: *Outturn
**Weighted by a share of Thailand's total exports to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)
***Weighted by a share of Thailand's total exports to major 13 trading partners as of 2014 (including the United Kingdom and Australia)
( ) reported in Monetary Policy Report June 2017

Most central banks maintained their accommodative monetary policy stances. However, some central banks raised their policy rates.

Most central banks maintained their accommodative monetary policy stances, although a few central banks raised their policy rates such as the Bank of Canada. As for the Federal Reserve (Fed), the Fed was expected to commence its balance sheet reduction[1/] in October 2017 and gradually raise the federal funds rate given the economy would be approaching full-employment amid the sluggish inflation outlook. However, central banks of some Asian economies with relatively high policy interest rates—namely, the State Bank of Vietnam, the Reserve Bank of India, and Bank Indonesia—cut their policy rates in the third quarter of this year.

Capital flows were highly volatile as a result of increased uncertainties from the conduct of monetary policy in advanced economies and geopolitical risks following the tensions in the Korean Peninsula.

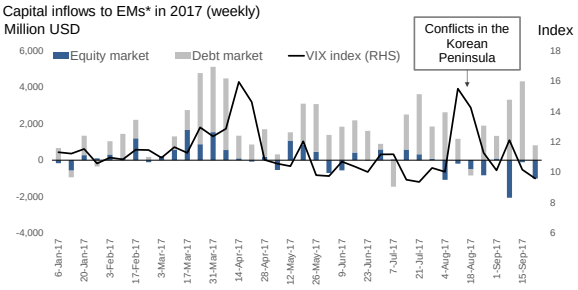
Toward the end of the second quarter of 2017, capital inflows to emerging markets (EMs) slowed down as the European Central Bank (ECB) and the Bank of England (BOE) tended to be more optimistic about their economies. As a result, investors anticipated monetary policy tightening in these countries to be earlier than previously expected. In July, capital inflows to EMs increased after U.S. economic figures turned out to be weaker than expected, especially inflation that still remained below the target. Moreover, some members of the Federal Open Market Committee reiterated their support for the gradual pace of policy rate normalization. Therefore, investors expected a lower likelihood of the Fed's rate hikes than previously anticipated and shifted their attention toward EMs assets especially into bond markets. Later at the end of August, capital outflows from EMs accelerated, particularly from equity market, due to heightened geopolitical risks on tensions in the Korean Peninsula. Amid the risk-off sentiment, investors adjusted their portfolios by reducing investment in EMs and instead increasing

[1/] The Fed would gradually reduce the size of its balance sheet according to the previously announced plan. The reinvestment caps would be 10 billion U.S. dollars per month for the first three months and would increase in steps of 10 billion U.S. dollars at three-month intervals over 12 months until it reaches 50 billion U.S. dollars per month. After that, reinvestment caps would be kept at 50 billion U.S. dollars per month until the balance sheet reaches an optimal level.

investment in low-risk assets. This was reflected by a surge in the Volatility Index (VIX)[2/] and prices of safe-haven assets. In the beginning of September, however, investors' concerns were declined and capital reversed to EMs, particularly in bond markets (Chart 1.3).

Looking ahead, financial markets would likely remain volatile. Fluctuations in capital flow movements, both into and out of Thailand, might result from uncertainties pertaining to U.S. economic policies, monetary policy directions of major advanced economies, and geopolitical risks.

Chart 1.3 Conflicts in the Korean Peninsula led to capital outflows from EMs especially from equity markets. However, after investors' concerns subcided, capital flows reversed particularly into bond markets.



Note: *EMs includes Thailand, Indonesia, South Africa, and Turkey

Sources: Bloomberg and Institutional Institute of Finance

The Dubai crude oil price in the third quarter of 2017 remained stable from the previous quarter, but would slowly rise underpinned by global economic recovery and production cuts by major oil producers.

The Dubai crude oil price in the third quarter of 2017 remained largely unchanged from the previous quarter. In July, oil prices dropped mainly due to supply-side factors as reflected in the large stock of global crude oil. Crude oil output rose for both groups of OPEC members: those under the agreement of production cuts and those exempt from the agreement, namely Libya and Nigeria. In August and September, however, oil prices gradually picked up after a sharp decline of oil stocks in the U.S. owing to more robust global economic recovery and the U.S. oil rig count that appeared to be stabilizing after having risen significantly earlier.

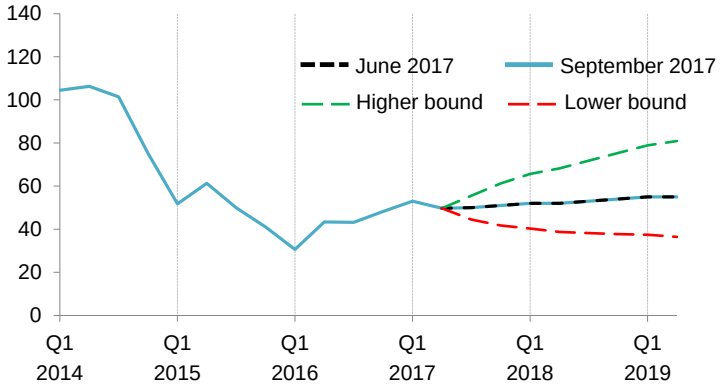
In the period ahead, oil prices would gradually trend up mainly on account of global demand recovery and the extension to the production cuts agreement between OPEC and Non-OPEC until the first quarter of 2018 which would steadily curb oil inventories**.** However, a continued expansion in shale oil production in the U.S. would restrain increases in oil prices going forward**.**

Against this backdrop, the Committee kept the projection for the Dubai oil price unchanged at 50.9 and 52.8 dollars per barrel in 2017 and 2018, respectively. Risks to the projection remained balanced. On the upside, geopolitical risks in the Korean Peninsula and the Middle East could help lift oil prices somewhat. Additionally, there were possibilities that the OPEC and non-OPEC countries might introduce new measures to stabilize prices and extend the cuts further. On the other hand, downside risks to the projection could stem from a higher-than-expected supply from those under the production cuts agreement, shale oil producers in the U.S., as well as Libya and Nigeria.

[2/] The VIX index indicates volatility of stock markets, calculated by implied volatility from the S&P 500 options.

Chart 1.4 Dubai oil price was largely unchaged from the previous quarter but was expected to gradually increase in the future.

Assumption on Dubai oil price
U.S. dollar/barrel



Metal prices in various types edged up in the short term such as iron ore, copper and aluminum, although both demand- and supply-side factors remained relatively unchanged. This was due to expectations on the outcome of the Chinese government's policy to curb production in four major provinces and policy to curb price speculation. Looking ahead, metal prices in 2018 would likely remain close to the previous assessment, despite greater downside risks from acceleration in China's production after having fallen in earlier periods.

2. The Thai Economy

2.1 Recent Developments

Thailand's economic growth in the second quarter of 2017 was stronger than the previous assessment in the *Monetary Policy Report* (June 2017) underpinned by expansion in merchandise exports and tourism and a continued recovery in domestic demand that began to be more broad-based. Meanwhile, public spending remained an important growth driver.

The Thai economy expanded 3.7 percent in the **second quarter of 2017** from the same period last year, an improvement from 3.3 percent in the previous quarter. The key growth driver was an accelerated expansion of merchandise exports across almost all product categories and export destinations. In particular, exports of rice and electronics expanded in tandem with improved external demand, the technological upcycle, and an expansion of production bases in Thailand. Exports of services accelerated in line with tourism receipts. The number of foreign tourists increased for all nationalities, especially Chinese tourists. Notably, the benefits of merchandise and services exports began to spill over to domestic demand. Private consumption continued to grow in line with spending on services and durable goods and rising consumer loan growth. Meanwhile, the public sector remained a key growth driver thanks to an acceleration in public consumption despite a contraction in public investment. Such decline was partly owing to expedited disbursements in the prior period on projects under the Highway Department, the Rural Highway Department, and the Irrigation Department. Some government agencies were also constrained by limited disbursement efficiency. However, private investment growth resumed after having declined in the previous periods, consistent with an expansion in machinery and equipment and construction. This helped offset the contraction in public investment and contributed to a slight increase in overall investment growth. Overall, the Thai economy recorded a 1.3 percent growth, after seasonal adjustment, in the second quarter of 2017, close to the growth rate of the previous quarter.

Thailand's economic growth continued to gain further traction **over the first two months of the third quarter.** Merchandise exports expanded both in terms of volume and price. This development was in line with regional exporting patterns and an economic recovery of Thailand's trading partners. Furthermore, a rise in export values was also evident across almost all product categories and export destinations with the benefits from the export expansion seen across firms of both large and small sizes. Consequently, rising employment in manufacturing sectors of certain industries was observed, which would provide growth momentum for private consumption going forward. Meanwhile, imports of machinery and equipment picked up in the manufacturing sector with improved exports, resulting in a higher private investment in the same category. Services exports continued to expand mainly on account of tourism receipts. Nevertheless, an expiration of measure regarding the reduction and exemption of tourist visa fees in August 2017 would yield limited impact on Chinese tourists. This was due to an increasing number of Chinese tourists in recent periods both quality tourists and free-independent travelers (FIT), whose spending power was higher than that of group tourists. Meanwhile, public spending expanded from both current and investment expenditure, particularly investment spending on state-owned enterprise investment projects during the initial phase of the Orange Line construction (Thailand Cultural Center–Minburi).

Income improved for nonagricultural households in export- oriented manufacturing and tourism sectors but overall household purchasing power was not yet robust.

Nonagricultural households benefited from higher employment and overtime payments in export- and tourism-related businesses such as food industries, rubber products, electronics, retail businesses, and hotels and restaurants. Such improvement was observed in higher employment across firms of all sizes, which helped lift purchasing power and provided a growth momentum for private consumption. On the other hand, employment in domestic-oriented sectors such as beverages, metals and metal products, and construction, was not evident. Therefore, employment outside the agricultural sector remained stable overall. With regard to income, income distribution improved in large and medium-sized businesses. However, the ongoing economic expansion in recent periods did not lend support to a better income distribution among workers in small businesses (BOX: *Distribution of growth dividends: evidence from the labor market*). Income of **agricultural households** picked up thanks to higher output of major crops such as rice and fruits. However, farm income was expected to slow down in the period ahead following a fall in agricultural prices that accompanied higher agricultural output. In addition, floods in some areas in the northeast would dampen agricultural households' confidence which would likely affect spending to some extent.

Regulations on immigrant workers might affect private consumption and inflation going forward.

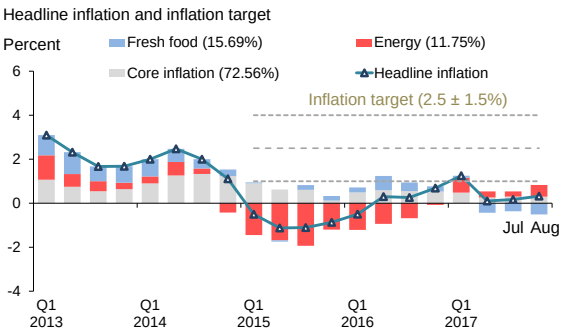
The government undertook measures to manage and legalize immigrant workers in Thailand through the issuance of the Royal Decree on Managing the Work of Aliens effective on June 23, 2017 where the reprieve period was extended to December 31, 2017 in order to allow time for business adjustments. Since then, majority of illegal immigrant workers in the agricultural and household sector had duly registered. Meanwhile, immigrant workers in the manufacturing and services sectors of large businesses were not much affected since most of them were legal workers. On the other hand, the numbers of immigrant worker registrations in small and medium-sized enterprises (SMEs) were still low. This was especially the case in hotels and restaurants, constructions, and trading businesses with a larger proportion of immigrant workers employed relative to other sectors. Therefore, these sectors could face risks of labor shortage in the period ahead. In addition, registered illegal workers must undergo nationality verifications which could take some time. The Committee viewed that the impact on certain businesses in risk sectors would be quite considerable in the short run, both on employment and economic activities. In addition, the Committee would monitor the impacts of these regulations on domestic consumption following potential losses of purchasing power of immigrant workers, and the impacts on the inflation outlook given a possible wage rise due to a tightening labor market, as well as a long-term impact such that these regulations might also result in an increase in costs of hiring immigrant workers.

Headline inflation edged up slightly due primarily to energy prices, while core inflation was stable relative to the previous quarter.

Headline inflation edged up slightly, averaging at 0.25 percent over the first two months of the third quarter, an improvement from 0.1 percent in the previous quarter (Chart 2.1). The increase was attributable to higher energy prices caused by (1) higher domestic retail oil prices in line with global crude prices and (2) higher energy charges (FT). Meanwhile, a decline in fresh food prices was less pronounced because the base effects of higher fresh food prices following

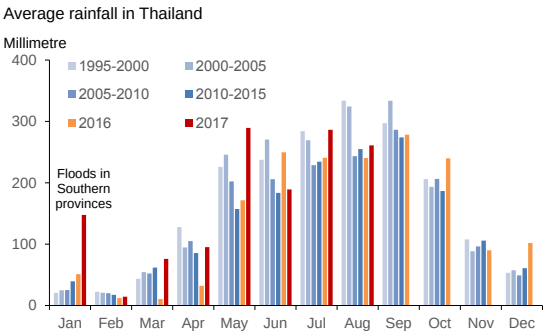
last year's drought dissipated. However, this year's weather conditions that were favorable for the output of vegetable and fruits remained a key factor behind a gradual rise in fresh food prices (Chart 2.2). In addition, five-year-ahead inflation expectations of professional forecasters fell slightly to 2.3 percent, which remained close to the midpoint of the target at 2.5 percent.

Chart 2.1 Headline inflation picked up slightly from the previous quarter mainly due to energy prices.



Note: ( ) denotes share in inflation baskets.
Source: Ministry of Commerce, calculations by Bank of Thailand

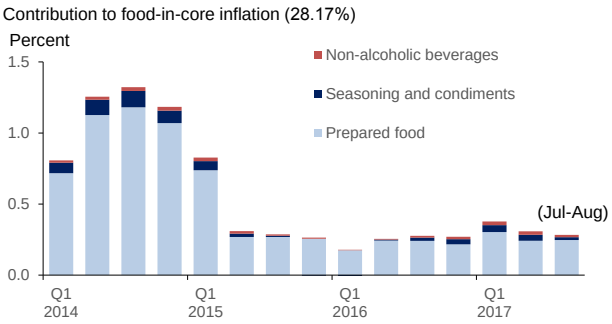
Chart 2.2 Favorable weather conditions, as reflected in higher than historical average rainfall, led to higher agricultural output and put downward pressures on fresh food prices.



Source: The Meteorological Department and the World Bank

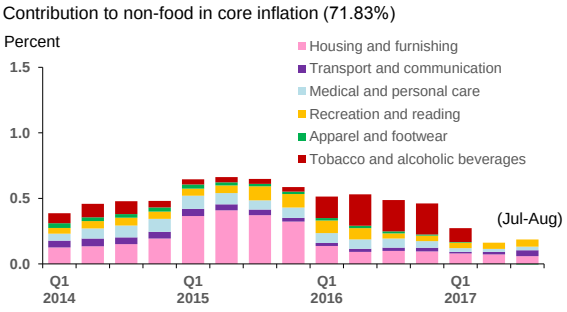
Core inflation was close to the rate of the previous quarter, averaging at 0.47 percent over the first two months of the third quarter. This was because prices of food items in the core inflation basket were stable at low levels owing mainly to lower costs of fresh food and liquefied petroleum gas (LPG) (Chart 2.3). Prices of nonfood components in core inflation were also stable at low levels in tandem with a gradual domestic demand recovery (Chart 2.4). In addition, structural factors including the rising trends of e-commerce and globalization, which enhanced competitions among both local and foreign businesses, and an expansion of global value chains, which enabled countries to gain easier access to cheaper raw materials, reduced the power of domestic firms in raising prices. Moreover, technological advancements also led to lower costs of production which partly put downward pressures on inflation and also weighed on an increase in prices of nonfood components in core inflation in the recent periods.

Chart 2.3 Food-in-core inflation remained low due to low LPG and fresh food prices.



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by the Bank of Thailand

Chart 2.4 Non food-in-core inflation remained low given little expansion of goods and services prices which was in line with slow expansion of domestic demand.



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by the Bank of Thailand

Table 2.1 Inflation

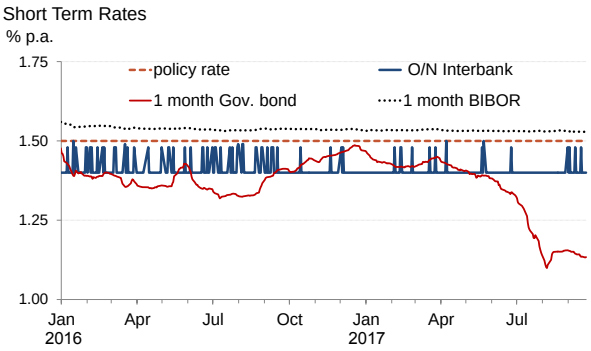
Annual percentage change	2016				2017		
	Q1	Q2	Q3	Q4	Q1	Q2	Jul-Aug
Headline Consumer Price Index (Headline CPI)	-0.50	0.30	0.26	0.69	1.25	0.10	0.25
Core Consumer Price Index (Core CPI)	0.67	0.79	0.76	0.73	0.66	0.47	0.47
Raw food	1.52	4.24	2.58	1.54	0.61	-2.97	-2.84
Energy	-11.41	-8.95	-7.00	-1.06	6.69	2.67	4.02

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Government bond yields declined across all tenures.

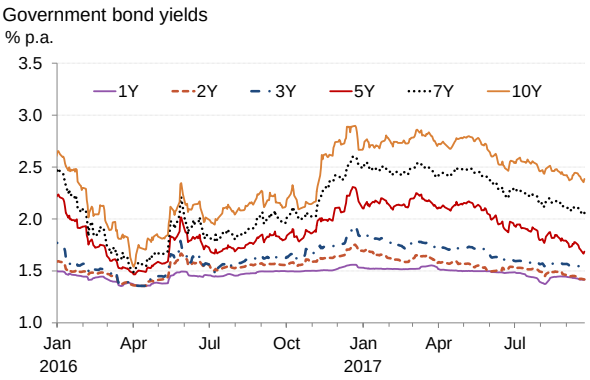
Short-term money market rates remained close to the policy interest rate in the third quarter of 2017. However, **short-term government bond yields** declined sharply in July (Chart 2.5) as a result of the reduction in the short-term bond issuances by the Bank of Thailand since April.[3/] During the same period, there was also a suspension of 28-day treasury bill issuances by the government at the end of the fiscal year 2017 and before the beginning of the next fiscal year.[4/] Nevertheless, short-term bond yields began to pick up since August due to market adjustments but were still lower than the policy rate. Meanwhile, **medium- and long-term government bond yields** continued to fall since July influenced by external factors. These included heightened political uncertainties in the U.S., which could result in a delayed implementation of domestic stimulus measures, and lower-than-expected U.S. economic data outturns especially inflation that was still below target. Investors, therefore, expected the increase in the federal funds rate to be slower than previously assessed. Moreover, rising tensions in the Korean Peninsula since the end of August also put pressures on yields to decline further (Chart 2.6). In September, bond yields of all tenures were below levels at the end of the previous quarter and at the end of 2016. **Corporate bond yields** declined alongside decreases in government bond yields and credit spreads. Meanwhile, **financing costs through commercial banks**, as reflected in the new loan rate (NLR)[5/], were stable at a low level after having declined continuously. (Chart 2.7).

Chart 2.5 Short-term government bond yileds dropped signicantly in July but edged up slightly while other short-term rates were largely unchanged.



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Chart 2.6 Medium- and long-term government bond yields trended down owing to external factors.



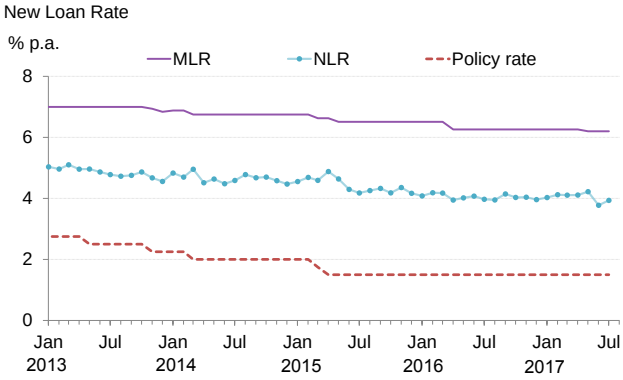
Sources: Thai Bond Market Association (Thai BMA)

3/ Since April 2017, the Bank of Thailand reduced the issuances of 3-month and 6-month bonds from 40 billion to 30 billion baht for each type.

4/ The government made adjustment to its treasury bill issuance plan in the fourth quarter of fiscal year 2017

5/ NLR is calculated based on a weighted average of interest rates for new loan contracts extended by 14 Thai commercial banks (excluding consumer loans and loans to financial intermediaries). The data covers loans of value of 20 million baht or higher for all purposes and terms, and includes both secured and non-secured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

Chart 2.7 New loan rate (NLR) was stable at low level after trending down in the previous period.



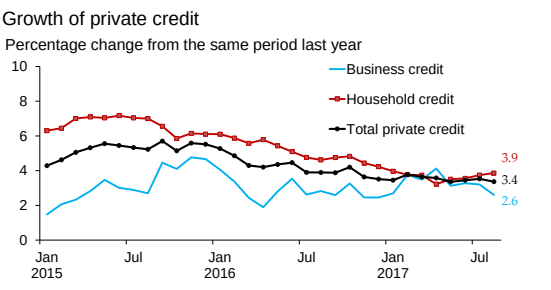
Source: Bank of Thailand

Private credit, particularly credit to households and large businesses in export-related and tourism sectors, continued to rise.

Private credit[6/] expanded 3.4 percent from the same period last year in the second quarter of 2017 (Chart 2.8). Commercial bank loans increased mainly on account of working capital loans to large corporates in export-related sectors. In August 2017, private credit grew 3.4 percent from the same period last year on the back of household credit growth, particularly from an acceleration of commercial bank lending in almost all loan purposes except credit card loans, which were largely unchanged. Auto loans continued to rise as the cars bought under the first-car scheme reached the five-year contract period, so demand for new cars increased. Meanwhile, mortgage loans extended by specialized financial institutions remained elevated. Business loan extension by commercial banks slowed down on account of debt repayments by large corporates. Nevertheless, working capital loans to large corporates in export-related and tourism sectors continued to expand, particularly in businesses related to rubber and plastic products, electronics, electrical appliances, food, and hotels.

Funding through the bond market during July–August 2017 slowed relative to the previous quarter as matured bonds of some telecommunications and energy companies were not replaced by new bond issuances. However, funding activities increased among businesses related to financial institutions, construction materials, petrochemical and chemical products (Chart 2.9). **Funding through the equity market** increased markedly in businesses related to food, banking, petrochemical and chemical products. The increase in funding was mainly for financial restructuring and business expansion purposes.

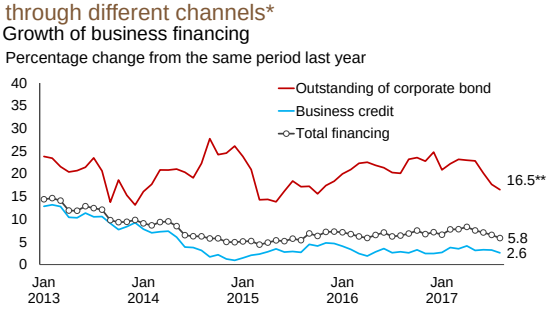
Chart 2.8 Growth in private credit increased mainly on account of household credit.



Note: Private credit includes credit to other depositary corporations (ODC) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds

Source: Bank of Thailand

Chart 2.9 Growth in business financing outstanding through different channels*

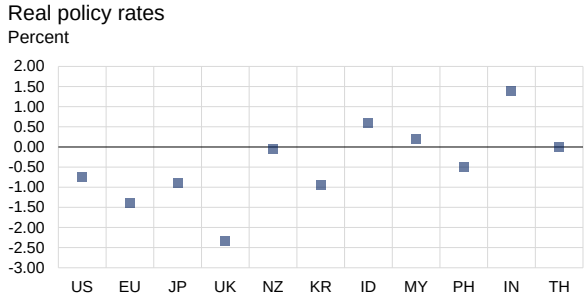


Note: *Business credit covers lending activities of Other Depository Corporations (ODCs) namely commercial banks, special financial institutions, saving cooperatives and money market mutual funds
**There was no rollovers of matured bonds of telecommunication and energy businesses in August, resulting in softer growth of the outstanding.

6/ Outstanding credit of other depositary corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and market mutual funds.

Going forward, financial conditions were expected to remain accommodative as reflected in the real policy rate which remained at a low level and was moderate compared with other countries (Chart 2.10), together with the Thai government bond yields that declined across all tenures as a result of a reduction in short-term bond supplies and external factors. Meanwhile, commercial banks' benchmark retail interest rates were stable at low levels. However, it was expected that financial institutions would still maintain caution in extending credit to households and certain businesses, particularly SMEs with deteriorating credit quality.

Chart 2.10 Thailand's real policy rate* remained low and was moderate relative to other countries.



Note: *Calculated from policy rate less 1-year ahead inflation expectation surveyed by the Consensus Forecasts (as of 11 September 2017)
Source: Consensus Economics, calculations by Bank of Thailand

The baht appreciated against the U.S. dollar, but the overall baht's movement relative to those of trading partners was largely unchanged.

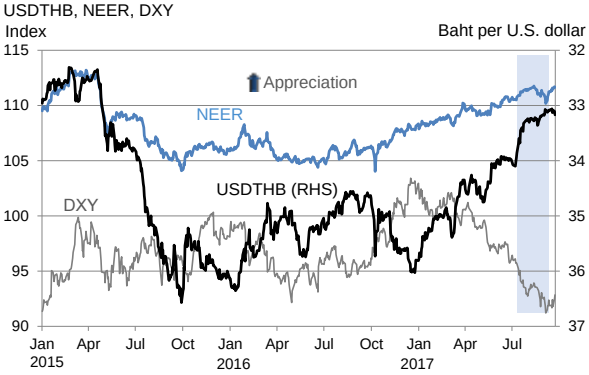
In the third quarter of 2017, **the baht strengthened against the U.S. dollar** relative to the end of the previous quarter. The baht appreciation was mainly due to the weakening of the U.S. dollar, (Chart 2.11) underpinned by negative market sentiments on the U.S. dollar from heightened political uncertainties in the U.S. which could delay the implementation of domestic economic stimulus measures and lower-than-expected U.S. economic data outturns especially inflation which was still below target. Markets, therefore, expected the increase in the federal funds rate to be slower than previously assessed. Furthermore, rising geopolitical risks from tensions in the Korean Peninsula also put additional pressures on the U.S. dollar.

In this quarter, sizeable capital inflows into regional stock markets including Thailand were observed. However, the baht recorded a stronger appreciation relative to most regional currencies in some periods especially in the latter half of July. This was on the back of a better-than-expected Thailand's economic data outturns, particularly a larger-than-projected current account surplus as reflected in persistent net sales of U.S. dollar by businesses. This was partly caused by an increase in gold exports at the time of rising geopolitical risks. Moreover, there was also a large amount of overseas fundraising by both Thai corporates and foreign affiliates. As of September 26, 2017 the baht closed at 33.17 to the U.S. dollar, appreciating 2.4 percent from the end of the previous quarter.

The nominal effective exchange rate (NEER) stood at 110.65 on September 26, 2017, appreciating 1.2 percent from the end of the previous quarter. Such appreciation was driven by the baht's appreciation against the U.S. dollar and regional currencies, but was partially offset by the baht's depreciation against the euro, pound sterling, and yuan. This was due to a euro-area economic data outturns which reflected a stronger growth outlook and the People's Bank

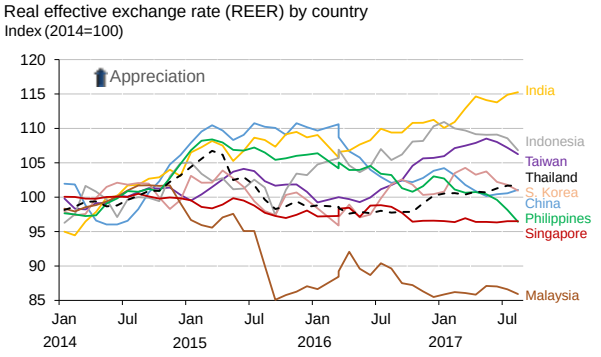
of China's revaluation of the yuan to be in line with better economic figures. As of the end of September 2017, the **real effective exchange rate (REER)** rose about 2 percent from the end of last year, but was less than the NEER's appreciation of roughly 3 percent. This was attributable to Thailand's relatively low inflation compared with trading partners which helped alleviate the impact of NEER's appreciation on price competitiveness of businesses to some extent. Moreover, Thailand's REER was moderate compared with regional peers and did not change significantly (Chart 2.12). Going forward, exchange rates would likely remain volatile due to external uncertainties such as uncertainties surrounding U.S. economic policies, monetary policy conducts of major advanced economies, and geopolitical risks.

Chart 2.11 The baht strengthened against the U.S. dollar as the U.S. dollar weakened across the board.



Source: Bank of Thailand and Reuters (data as of 26 September 2017)

Chart 2.12 Thailand's real effective exchange rate appreciated rather moderately in comparison to other regional economies.



Source: BIS, calculations by the Bank of Thailand (data as of September 2017)

Financial stability remained sound overall. However, there remained pockets of risks that warranted monitoring, particularly debt serviceability of households and SMEs as well as a search-for-yield behavior that might lead to the buildup of vulnerabilities in the financial system in the period ahead.

Thailand's financial stability remained sound overall with strong external positions as reflected in a high level of foreign exchange reserves and sustained current account surpluses which would provide cushion against volatilities in global financial markets. Meanwhile, financial positions of large corporates and financial institutions provided a positive outlook, given that financial institutions maintained high levels of capital buffers and provisions for loan losses to cushion against risks stemming from deteriorating credit quality. Nevertheless, the Committee assessed that there remained pockets of risks that could potentially lead to the buildup of vulnerabilities in the financial system in the period ahead and thus warranted close monitoring. These risks are summarized as follows.

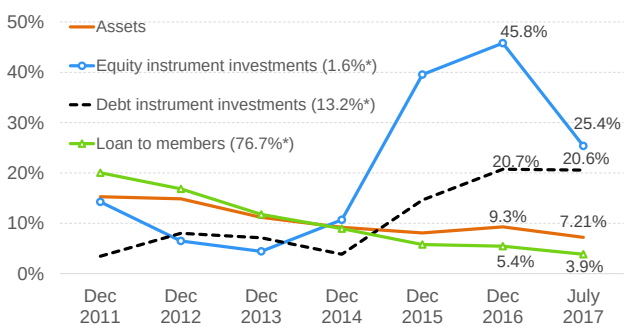
(1) The continuation of the search-for-yield behavior in the prolonged low interest rate environment. Although overall systemic risks were largely limited, certain risks that indicated signs of underpricing of risks must be monitored. It was evident that investment in assets that were riskier than domestic deposits continued to increase. This was especially the case for foreign investment funds (FIF) which mainly invested in deposits and short-term debt securities in countries with investment-grade credit ratings, but there was still risk of investment concentration in some countries. Meanwhile, investment in funds with deposit-like characteristics was stable relative to the previous quarter. In addition, investment in unrated bonds trended down following defaults of some issuers at the end of 2016 as issuers of those defaulted bonds

switched to borrowing from commercial banks. Meanwhile, investors were more cautious, while risk assessment capability improved as reflected in higher funding costs of issuers with fragile financial positions. Similarly, asset management companies exercised greater caution when investing in unrated bonds.

Furthermore, deposits at saving cooperatives, both in the form of money deposits and investments in cooperative stocks, continued to expand. Given that deposit rates and average dividends of saving cooperatives were higher than those of commercial banks, some saving cooperatives accepted deposits from members in the amount exceeding loan demand. Such situation might prompt saving cooperatives to seek higher yields, both in the forms of investment in risky assets or lending to high-risk borrowers, especially to certain cooperatives. However, investment in equities of saving cooperatives began to slow somewhat (Chart 2.13). The Committee would continue to closely monitor these developments as there appeared to be growing linkages between saving cooperatives and financial system by way of specialized financial institutions and commercial banks.

Chart 2.13 Investments in equity instruments of saving cooperatives slowed down.

Asset growth and investment growth of saving cooperatives



Note : *share to total assets of saving cooperatives (as of December 2016)

Source: Cooperative Auditing Department, calculations by Bank of Thailand

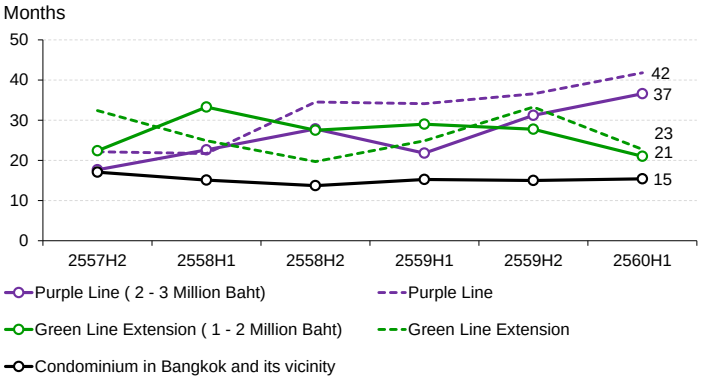
(2) Debt serviceability of households and SMEs. Ability to service debt remained fragile as reflected in a high ratio of nonperforming loans (NPL) to overall consumer loans, despite some improvements in the second quarter of 2017. However, share of NPLs in housing loans continued to increase, which was partly a result of financial institutions' competitions in credit extension during 2015–2016 following the government's real estate stimulus measures. Moreover, interest rates that remained low for an extended period might discourage savings, which could affect national savings and increase households' vulnerabilities to economic fluctuations. Meanwhile, debt serviceability of the business sector, especially SMEs, continued to deteriorate. This was largely owing to specific problems in the industrial, commercial, and construction businesses that were traditionally operated with limited capital. Thus, these businesses faced constraints in adapting themselves to competitions and a rapidly changing business environment. Nevertheless, quality of some business loans improved thanks to export growth. Meanwhile, situation regarding the rollover of the unrated corporate bonds also improved. However, unrated bonds that would mature in the period ahead still warranted monitoring, especially those of issuers with fragile financial positions as well as some bonds with B-credit rating where defaults began to emerge.

(3) Oversupply of condominiums in Bangkok and its vicinity. Although overall risks were stable, there remained the need to monitor the situation regarding an increasing number of unsold units of condominiums in certain areas and price ranges and a longer "time to go," especially for condominium units along the MRT Purple Line (Chart 2.14). Developments regarding the launches of large-scale mixed-used real estate projects must also be monitored, particularly some recently launched projects that were expected to be gradually completed and open for sale from 2018 onward. Moreover, there remained risks that warranted monitoring

regarding increasing risks of maturity mismatches between assets and liabilities of some real estate developers amid growing excess supply in the condominium market.

Chart 2.14 Condominiums in certain areas and price ranges saw longer "time to go"

Average "time to go" of condominiums along the mass transit system route



Note : "Time to go" is the time taken for all real estate inventory to be sold, using average sales rate of the past 12 months.

Source: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

2.2 Outlook for the Thai economy

Under the Committee's assessment, Thailand's economic growth was projected to gain further traction in the period ahead and attain a higher growth rate of 3.8 percent in both 2017 and 2018, compared with the assessment in the previous *Monetary Policy Report*. The main drivers of growth were expected to be from both external and domestic factors. These included (1) improvements in merchandise exports and tourism in line with a stronger growth outlook of Thailand's trading partners, (2) a gradual rise in private spending that began to be more broad-based, and (3) a sustained fiscal impetus despite unexpected delays in some public investment projects. Meanwhile, inflation was projected to gradually edge up overall albeit at a slower pace than previously assessed due mainly to supply-side factors.

Table 2.2 Summary of forecasts

Percent	2016*	2017	2018
GDP growth	3.2	3.8 (3.5)	3.8 (3.7)
Headline inflation	0.2	0.6 (0.8)	1.2 (1.6)
Core inflation	0.7	0.6 (0.6)	0.9 (0.9)

Note: * Outturn
() Monetary Policy Report June 2017
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of key forecast assumptions

- Trading partner economies were projected to achieve higher growth rates than previously assessed as most growth outturns in the second quarter of 2017 were better than expected. In particular, Asian economies continued to grow on the back of exports and private consumption.
- The federal funds rate was expected to be raised one more time over the remainder of 2017, while it was also expected that the Fed would start reducing its balance sheet in accordance with the announced plan. In 2018, the Fed would likely raise the policy interest rate three times.
- Asian currencies (excluding the Chinese yuan) were stronger than the previous assessment given their appreciation in the third quarter of 2017. In the period ahead, however, the currencies were projected to depreciate following increases in the federal funds rate.
- The Dubai crude oil price was projected to be the same as the forecast in the previous *Monetary Policy Report* due to underlying fundamentals that were largely unchanged.
- Metal prices were expected to increase in the short run due to a plan to cut metal production announced by the Chinese government. Going forward, metal prices were projected to remain close to the previous projection given that underlying fundamentals remained largely unchanged.
- Farm income was revised down from the previous assessment due to lower agricultural prices which was a result of a higher-than-expected agricultural output, especially palm oil and fruits, thanks to favorable weather conditions. Nevertheless, a higher agricultural output could only partially offset a fall in agricultural prices.
- Public spending at current prices was revised down due to unexpected delays in public investment especially from a lower-than-expected disbursement of investments by central government and a reduction in the budget of state-owned enterprises in 2018. Meanwhile, public consumption was also revised down slightly due to delayed disbursement of additional budget in 2017, but was expected to rise in 2018 in accordance with a higher proportion of current spending in the 2018 fiscal year budget.

Table: Summary of forecast assumptions

Annual percentage change	2016*	2017	2018
Dubai crude oil price (U.S. dollar per barrel)	41.4	50.9 (50.9)	52.8 (52.8)
Metal prices (% YoY)	-5.3	20.8 (18.6)	-0.5 (0.5)
Farm income (% YoY)	1.6	6.3 (6.5)	4.3 (4.4)
Government consumption at current price (billion baht)[1/]	2,456	2,566 (2,572)	2,710 (2,688)
Public investment at current price (billion baht)[1/]	936	987 (1,034)	1,123 (1,163)
Fed funds rate (% at year end)	0.63	1.38 (1.38)	2.13 (2.13)
Trading partners' GDP growth (% YoY)[2/]	3.1	3.6 (3.4)	3.4 (3.3)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	154.8	156.3 (157.6)	156.4 (159.8)

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report June 2017

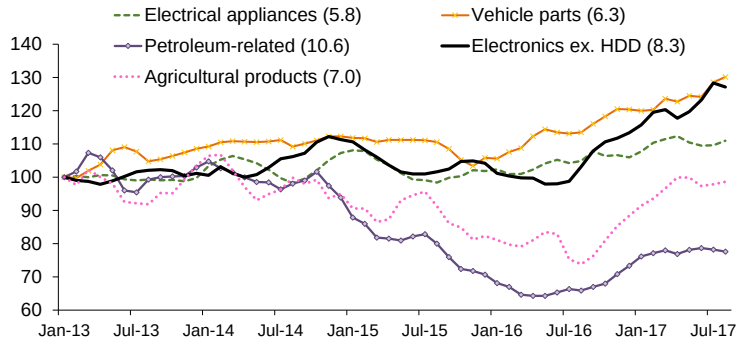
Merchandise exports were projected to continue expanding and become more broad-based.

The value of merchandise exports was expected to exhibit higher growth than those reported in the previous *Monetary Policy Report* and register 8.0 and 3.2 percent in 2017 and 2018, respectively. A stronger and continued expansion of the global economy and global trade contributed to a continued growth of Thai merchandise exports across various product categories such as electronics, electrical appliances, automobile parts, and agricultural products (Chart 2.15)**.** Growth was also expanded across almost all export destinations including developed markets, Asian countries, and the Middle East**.** Notably, Thai exports of electronics were expected to record further growth, especially integrated circuits which would be used as parts in Internet of Things (IoT) devices and electronic parts in automobiles**.** Exports of smartphones and motorcycles also benefited from the relocation of production bases to Thailand**.** Moreover, export prices were projected to rise in tandem with crude oil prices, especially for commodities, for example, rubber and oil-related goods such as petroleum products**.**

However, in the latter half of 2017 and throughout 2018, the growth outlook of Thai exports was projected to expand at a slower pace, compared with the first half of 2017. This was partly a result of high base effect in the previous year from, for instance, the relocation of production bases of automobile tires and hard-disk drives to Thailand. Also, export growth of particular products slowed down, including rubber exports to China given an already high level of rubber stocks and rice exports after prior accelerations in the previous period. Concurrently, there were also structural problems in certain Thai industries that would take time to be resolved. In addition, global trade in 2018 was projected to slow down after having accelerated this year. Looking ahead, key risks to the Thai export outlook included uncertainties surrounding the U.S. foreign trade policy, which could affect global economic growth outlook, and China's potential economic slowdown as a result of ongoing economic structural reforms.

Chart 2.15 Exports of goods exhibited a stronger recovery across many product categories, especially electronics

Value of exported goods
Seasonally adjusted index, 3-month moving average
(January 2013 = 100)



Note: Number indicated in () denotes share to total exports in 2016
Source: Customs Department, calculation by Bank of Thailand.

Exports of services were projected to gain further traction.

Exports of services were projected to expand further on account of continued improvements in the tourism sector due to several reasons: (1) an increasing number of Chinese tourists—both group and independent tourists with high purchasing power—thanks to the opening of new direct flight routes from China to major tourist destinations in Thailand, (2) a rising number of ASEAN tourists in line with regional economic recovery, and (3) the reduction and exemption of tourist visa fees which yield benefit to increasing numbers of foreign tourists. Although such measures already expired in August 2017, the impact on the overall number of tourists was relatively limited. Moreover, improvements in the global economy would be a supporting factor for an increase in the number of tourists from other countries and a further rise in tourism spending per head. Thus, **the Committee revised up the projection for the number of foreign tourists to 35.6 and 37.3 million in 2017 and 2018, respectively.**

Given improvements in the value of merchandise and services exports, the projection for the value of merchandise and services imports was revised up with higher imports of raw materials and intermediate goods. As a result, **the current account would continue to record large surpluses and would be higher than the previous estimates, registering surpluses of 42.4 and 38.6 billion U.S. dollars in 2017 and 2018, respectively.**

Private consumption was projected to expand at a gradual pace.

Private consumption was projected to gradually expand in the period ahead, supported by improvements in farm income due to higher agricultural output**.** However, falling agricultural prices would weigh on confidence and spending of agricultural households. Meanwhile, employment picked up in export-oriented manufacturing and tourism sectors, which helped shore up confidence of households outside the agricultural sector regarding their future income**.** In addition, private consumption benefited from several government measures, including the social welfare card project to support low-income individuals, the 9101 project[7/] and flood relief measures**.** Nonetheless, purchasing power would remain modest in the period ahead because employment and income had yet to fully benefit from the improved exports,

[7/] The 9101 project is a project for sustainable agricultural development in honor of His Majesty the late King.

partly due to an increasing adoption of automation in place of human labor as well as economic structural changes and business models that were less reliant on labor. Moreover, despite decreasing somewhat, elevated household debt remained a drag to consumption. Households would need some time to reduce their financial vulnerabilities (BOX: *Household debt deleveraging and implications for the economy*).

The public sector continued to drive the economy as budget disbursement was well on track despite some delay in certain investment projects.

Public spending remained an important economic growth driver, supported by budget disbursement that was well on track despite some unexpected delay in certain projects. Most investment projects by state-owned enterprises (SOE) continued as planned, except for the Bangsue–Huamak and Bangsue–Hualampong missing links under the suburban train projects that were still under consideration regarding construction plan in order to connect with the Airport Link extension. Furthermore, the budget for SOE investments in 2018 was also reduced. With regard to public investment, growth was expected to slow down after prior acceleration, while some government agencies were constrained by limited disbursement efficiency. In addition, the promulgation of the Public Procurement and Supplies Management Act, B.E. 2560[8/] might result in a delayed disbursement during the initial phase for the goods and services purchases and investments of some state agencies that would now be governed by the new law, particularly local administrations that had not previously operated under this system. Meanwhile, central government agencies were already accustomed to such system.

Private investment was projected to gradually recover.

Private investment continued to recover albeit at a slower pace. In the near term, investment recovery was observed in more industries, consistent with a stronger expansion of private consumption and exports**.** However, certain businesses still slowed down their investment, as there remained excess production capacity, and were still awaiting for greater clarity in the government's stimulus policies especially those regarding some infrastructure investment projects. Nevertheless, government policies were expected to play a key role in supporting private investment in the period ahead, both through a continuation of public investment and the future implementation of the Eastern Economic Corridor Act which would help shore up business confidence and attract greater foreign investment.

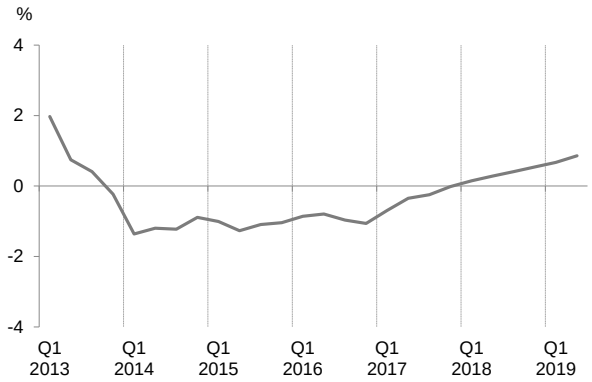
Inflation stabilized at a low level in the near term due to supply-side factors but was expected to slowly rise going forward.

In the recent periods, inflation remained at a lower level than previously assessed in the previous *Monetary Policy Report* due to a more pronounced impact from supply-side factors. In particular, such decline was driven by a fall in fresh food prices as a result of higher agricultural output of vegetable and fruits amid favorable weather conditions, together with last year's high base following the drought. Moreover, although external sectors were a key economic growth

[8/] The Public Procurement and Supplies Management Act, B.E. 2560 was an upgrade of the Regulations of the Office of the Prime Minister on Procurement, B.E. 2535 and the Regulations of the Prime Minister on Electronic Procurement, B.E. 2549 to be an Act. The Act enforces all state agencies to come under the same standards in order to increase efficiency and transparency of procurement process.

driver, domestic demand expanded in a gradual manner. Consequently, domestic prices were not expected to accelerate markedly.

Chart 2.16 Output Gap



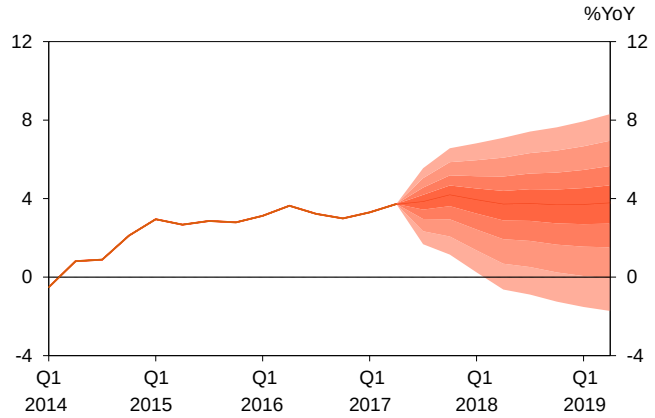
Going forward, inflation was expected to slowly edge up as **demand-pull pressures** gradually rose in tandem with the improved growth outlook of the economy. This was reflected in the closing of the output gap in the first half of 2018 (Chart 2.16). Moreover, upcoming government measures would further support household purchasing power. However, benefits from economic expansion were not yet widespread, while most businesses continued to offer sales promotion. As a result, demand-pull inflationary pressures were expected to be limited. On the other hand, **cost-push pressures** were expected to rise in accordance with an increase in excise tax on liquor, beer, tobacco, and sugary drinks as well as rising wages due to regulations on immigrant workers and a minimum wage rise in 2018. Nonetheless, there remained pressures that would cause inflation to rise at a slower pace than previously assessed. These included prices of fresh food that were expected to remain low, structural changes such as e-commerce and globalization that intensified competitions among businesses, and technological advancements that helped reduce costs of goods and services. **The Committee, therefore, revised down the projection of headline inflation to 0.6 and 1.2 percent in 2017 and 2018, respectively, while maintaining the projection of core inflation at 0.6 and 0.9 percent in 2017 and 2018, respectively.**

Risks to the growth forecast became more balanced in the near term, but remained skewed to the downside. Meanwhile, risks to inflation were judged to be balanced.

Under the Committee's assessment, the latest growth projection was characterized by a smaller degree of uncertainties, while risks to growth became more balanced over the remainder of 2017, albeit remaining tilted to the downside as displayed in the fan chart (Chart 2.17). Uncertainties surrounding the growth forecast decreased due to a stronger growth outlook of Thailand's trading partners. On the downside, there were possibilities that growth outturns might be lower than the baseline projection due to several reasons. First, the U.S. foreign trade policies might affect global economic growth outlook and lead to a slowdown in Thailand's trading partners' economies. Second, China's ongoing economic structural reforms could decelerate growth of China and its trading partners. Third, geopolitical risks might impact Thailand's trading partners' economies. In addition, there were also domestic risks which included (1) overall purchasing power that remained modest especially among low-wage earners whose income had yet to fully recover, (2) regulations on immigrant workers that could dampen economic activities more than previously anticipated, and (3) a possible slowdown in public spending from a more-than-expected impact of the Public Procurement Act. On the upside, there were possibilities that growth could outperform the baseline projection given a stronger-than-expected growth of the U.S. economy on the back of domestic stimulus measures which would further enhance growth of Thailand's trading partners. Nevertheless, **the Committee assessed risks to both headline and core inflation forecasts to be in balance** (Charts 2.18 and 2.19). On the downside, possibilities that inflation might fall below the baseline

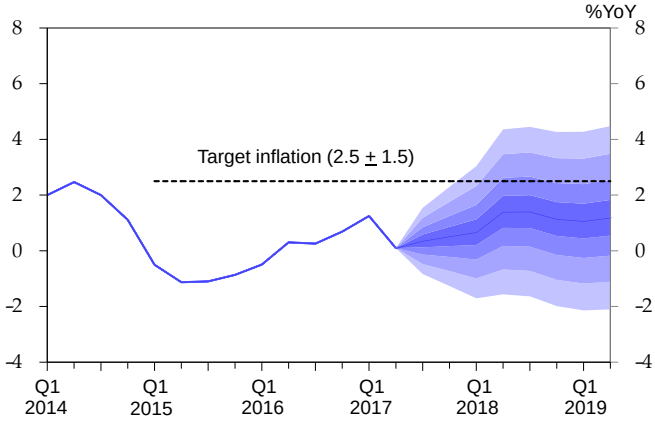
projection were from risks that growth would be lower than previously assessed and the impact of structural changes from technological advancements and intensifying business competitions. On the upside, inflation might be higher than the baseline projection given that regulations on immigrant workers resulted in a tightening labor market and would in turn push labor costs higher than previously projected. The extent to which labor costs would rise, however, would depend on the implementation and management by the authorities in the period ahead.

Chart 2.17 Growth forecast



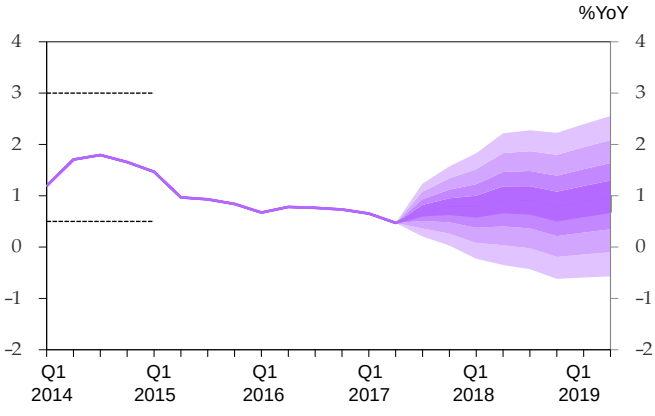
Note: Fan chart covers 90% of the probability distribution

Chart 2.18 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.19 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2016*	2017	2018
GDP growth	3.2	3.8 (3.5)	3.8 (3.7)
Domestic demand	2.7	3.0 (3.0)	3.5 (3.4)
Private consumption	3.1	3.3 (3.1)	3.0 (3.1)
Private investment	0.4	2.3 (1.7)	3.0 (3.6)
Government consumption	1.7	2.1 (2.2)	2.7 (1.9)
Public investment	9.9	5.0 (7.7)	9.8 (9.2)
Exports of goods and services	2.1	5.9 (4.6)	3.3 (2.7)
imports of goods and services	-1.4	6.5 (4.8)	3.3 (2.6)
Current account (billion, U.S. dollars)	47.7	42.4 (39.7)	38.6 (32.7)
Value of merchandise exports	0.1	8.0 (5.0)	3.2 (1.7)
Value of merchandise imports	-5.1	14.0 (10.9)	6.3 (5.4)
Number of foreign tourists (million person)	32.6	35.6 (34.9)	37.3 (37.3)

Note: *Outturns

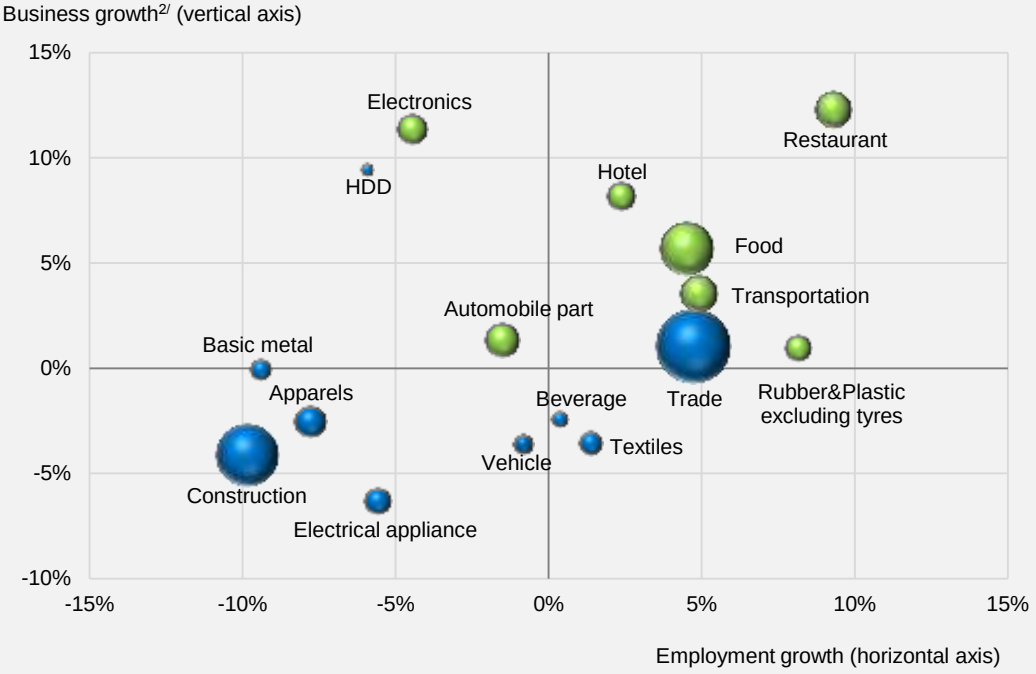
( ) Monetary Policy Report June 2017

Distribution of growth dividends: evidence from the labor market

The Thai economy continued to expand on the back of a stronger global expansion and started to have positive spillovers to the labor market. While overall **nonagricultural employment**[9/] remained stable during the first seven months of this year, there was higher employment in some certain businesses that benefited from exports and tourism sectors—namely, transportation, hotel and restaurant, processed food, and rubber and plastic (Chart 1). In addition, **these improved employment situation started to be increasingly observed in large and small businesses**, having witnessed earlier this year in medium-sized enterprises (Chart 2).

Chart 1 Employment in businesses that benefited from growth in exports and tourism increased.

Nonagricultural business growth and employment growth[1/]



Note: [1/] Business growth and employment growth were the comparison between the average over Janauary–July 2016 and the average over January–July 2017.

[2/] Business growth was calculated from the manufacturing production index (for manufacturing sectors), sales derived from VAT (for service sectors), and the construction index (for the construction sector).

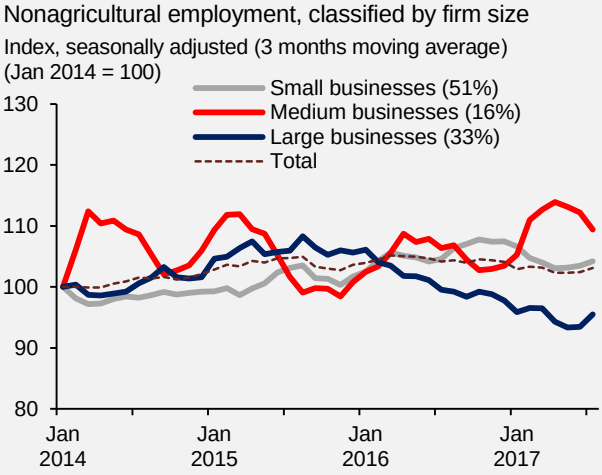
[3/] The bubble size represents that sector's share of employment in the overall manufacturing sector employment (for manufacturing sectors) or in the overall service sector employment (for service sectors).

[4/] Green bubbles represent the sectors that benefited from exports and tourism.

Sources: The National Statistical Office, the Office of Industrial Economics, the Revenue Department, and the Bank of Thailand; calculations by the Bank of Thailand

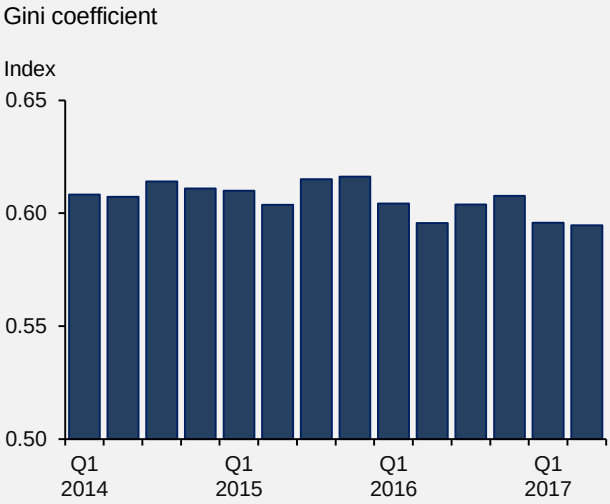
[9/] Nonagricultural employment refers to workers of private businesses in nonagricultural sectors, excluding workers of government entities.

Chart 2 Employment in medium-sized businesses increased since the beginning of the year, whereas employment in large and small businesses increased later



Note: ( ) denotes share in total employment
Source: The National Statistical Office, calculations by the Bank of Thailand

Chart 3 Income distribution of nonagricultural workers overall remained stable

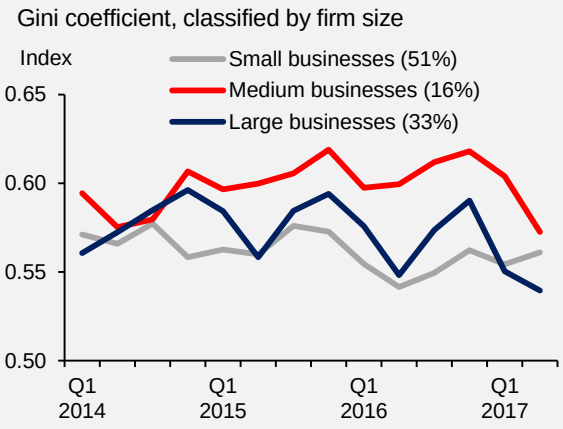


Note: The Gini coefficient approaching 0 represents an improvement in income distribution.
Source: The National Statistical Office, calculations by the Bank

In addition to the positive spillovers from overall economic growth to employment, this article examines whether and to what extent the recent economic growth was broad-based by considering (1) income distribution of nonagricultural workers based on Gini coefficients by sectors, and (2) nonagricultural employment by income groups. The findings are summarized as follows.

(1) The income distribution of nonagricultural workers overall did not yet improve, especially for small enterprises, but started to show signs of improvement among medium-sized and large businesses. This was reflected by the Gini[10/] coefficient of nonagricultural workers that remained stable during the first half of the year (Chart 3). However, when classified by firm size, income distribution of nonagricultural workers in medium-sized and large businesses was seen to have improved, while income distribution in small businesses slightly worsened (Chart 4). Based on this finding, income distribution of workers in small businesses had yet to benefit from the recent economic growth.

Chart 4 Income distribution of workers in large and medium-sized businesses improved whereas in small businesses deteriorated.

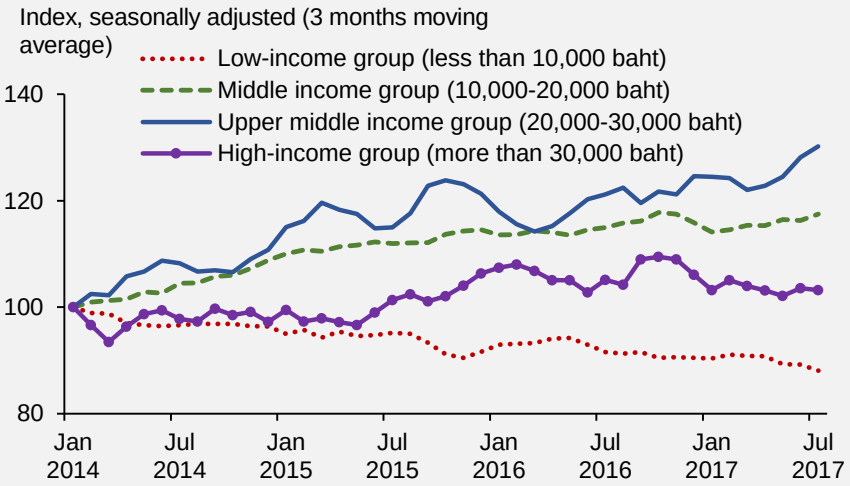


Note: ( ) denotes share in total employment
Source: The National Statistical Office, calculations by the Bank of Thailand

[10/] The Gini coefficient reflects income distribution and ranges from zero to one. A smaller Gini coefficient means a low-income group holds a larger share of a country's total income. As a result, there is less income inequality as well as a better income distribution.

(2) Nonagricultural employment in a middle-income group (10,000–30,000 baht per month) increased, while employment in high- and low-income groups remained mostly unchanged or moderated (Chart 5). Therefore, the decline in the Gini coefficient of large and medium-sized businesses as shown above could be due to higher employment in the middle-income group. Meanwhile, employment in the high-income group declined somewhat after having risen in earlier periods. Overall income distribution, however, did not improve much because employment in the low-income group had yet to recover. Moreover, impacts on economic growth of employment in the high-income group that was expected to decline warranted monitoring going forward.

Chart 5 Nonagricultural employment in middle and upper-middle income groups increased whereas employment in the low-income group had not yet recover.



Note: ( ) denotes the monthly income of worker group

Source: The National Statistical Office, calculations by the Bank of Thailand

In sum, **evidence from the labor market shows that economic growth was not yet sufficiently strong and overall income distribution had yet to improve**. On the one hand, employment increased in businesses that benefited from exports and tourism and started to be more broad-based as seen in various business sizes. Income distribution of workers in large and medium-sized businesses started to show some improvements. On the other hand, **employment in the low income group and income distribution among workers in small enterprises had yet to edge up**. Therefore, such developments in the labor market warranted close monitoring in order to ensure that the gains from economic growth could be distributed to various stakeholders in terms of both quantity and quality.

Household debt deleveraging and implications for the economy

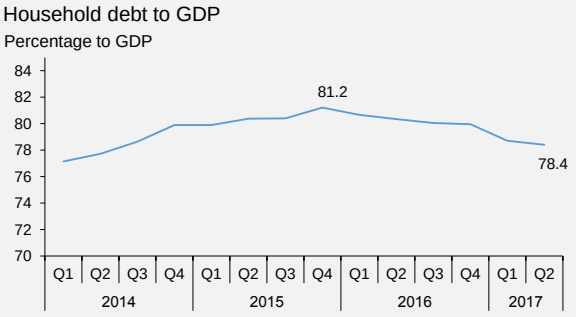
Household debt received considerable attention from the public because the ratio of household debt to GDP accelerated significantly since 2011. The reasons included the first-time car buyer scheme, the great flood during 2011–2012, and a surge in agricultural prices due to both global price cycles and government subsidies which raised farmers' income expectations. As debt accumulation outpaced income growth, debt serviceability of households continued on its downward trajectory as reflected in higher NPL ratios. Nevertheless, household debt growth had leveled off and continued to slow down, thereby resulting in a fall in the debt-to-GDP ratio for six consecutive quarters since early 2016. This situation was referred to as **"debt deleveraging,"** that is, a reduction in debt relative to income.[11/]

Debt deleveraging by Thai households

When household debt accumulation relative to income accelerates to certain levels, households' leverage will likely be limited as they are constrained by ability to service debt. Households, therefore, have to adjust their financial positions which marks the beginning of debt deleveraging. However, such situation can take place in various forms. One of the cases is through debt reduction, which usually occurs following an economic crisis originating from asset price bubbles in the real estate sector or other financial stability issues. As a result, this leads to a pronounced decline in credit extension and an extreme debt deleveraging. Another case is when loan growth is outpaced by income growth[12/]. This is generally the case when the economy is growing at a slower pace than normal, leading to a relatively moderate debt deleveraging.

In Thailand, deleveraging slowly took place since the beginning of 2016, with loan growth increasing more slowly than that of income. The latest data in the second quarter of 2017 revealed that the ratio of household debt to GDP fell to 78.4 percent (Chart 1)[13/]. However, deleveraging remained concentrated in certain income groups. According to the Household

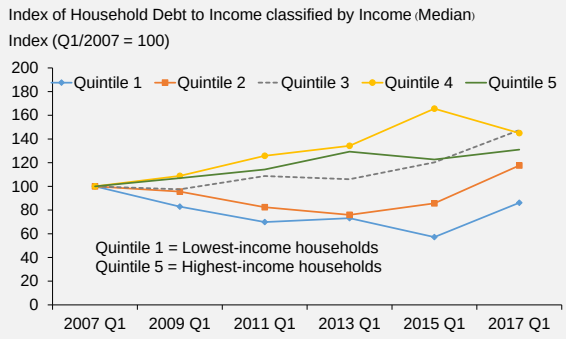
Chart 1 Household debt* to GDP was declining since the beginning of 2016



*Notes: Household debt does not include debt under litigation and credits under the Student Loan Fund

Source: Bank of Thailand

Chart 2 Signs of deleveraging were seen in high-income households (Quintile 4).



Source: Socio-Economic Survey, National Statistical Office; calculations by Bank of Thailand

11/ Many analysts use the debt-to-GDP ratio in place of the debt-to-income ratio because GDP data are comparable across countries and available in advance of household disposable income data. However, the use of GDP instead of household disposable income might not reflect households' financial adjustments, especially when the ratio of household disposable income to GDP changes significantly.

12/ Deleveraging occurs whenever loan growth is outpaced by income growth—regardless of whether loan growth is faster, stable, or slower than in the past.

13/ The ratio of household debt to GDP does not include debt under litigation and credit under the Student Loan Fund. If included, the ratio would be higher and the increase would be a parallel shift. Therefore, a deleveraging trend could still be observed.

Socio-economic Survey[14/] (Chart 2), **deleveraging was evident among some high-income households** (fourth quintile in Chart 2). This was primarily due to a decrease in debt while income was broadly unchanged. Meanwhile, low-income households continued to leverage further since 2015 owing to debt acceleration while income was largely unchanged or increased slightly. In addition, according to data reported to the Bank of Thailand by financial institutions, deleveraging was particularly apparent in auto loans given the end of loan contract terms under the first-time car buyer scheme. Housing and business loans exhibited a relatively slow deleveraging pace because such loans still continued to expand.

Implications of deleveraging for the economy

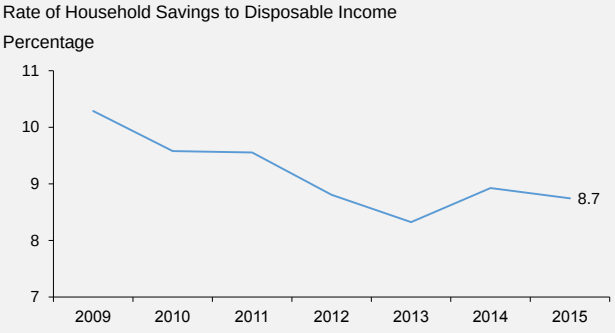
Household debt accumulation for consumption was equivalent to borrowing future income for today's consumption, which would affect consumption at a later date (Table 1). During the leveraging period, consumption growth would outpace income growth, leading to lower household savings. This behavior could not indeed last forever as households would be unable to additionally leverage while also servicing existing debts. Thus, households must adjust themselves by lowering consumption and saving more in order to pay back their debt. Therefore, during the deleveraging period, consumption would grow at a slower pace than income[15/]. However, a gradual deleveraging process following a debt acceleration would yield benefit in the long run because households' adjustment would alleviate financial vulnerabilities and strengthen their balance sheets. As a result, risks to financial stability would reduce and consumption would once again become an important economic growth driver.

Table 1 Pattern of Savings and Consumption in Different Periods

Variables	Leveraging Period	Deleveraging Period
Debt to Income	Increase ↑	Decrease ↓
Saving Rate	Decrease ↓	Increase ↑
Consumption	Increase faster than income	Increase slower than income

In the case of Thailand, the ratio of household savings to income was on a downward trend in tandem with debt accumulation (Chart 3). During 2008–2015, household consumption grew by 5 percent, a higher rate than household income growth of 4.7 percent. However, since 2017, as households began to deleverage, saving rates were expected to increase and thus consumption would likely grow at a slower pace than income. Nevertheless, during the deleveraging period, an appropriate level of monetary policy accommodation would play an important part in facilitating households' adjustments, reducing debt burden by lowering financing costs and supporting an increase in household income. It would also improve efficiency in household debt restructuring which would lead to an orderly deleveraging, facilitating household consumption to resume its role in supporting stronger economic growth going forward.

Chart 3: The ratio of household savings to income trended down during the leveraging period.



Source: Office of the National Economic and Social Development Board

[14/] This quarterly data were collected by the National Statistical Office based on a sample size of 10,000 households. In Charts 2 and 3, data were from 2007Q1 to 2016Q1. Income data were available every other year.

[15/] During deleveraging in some case, savings did not increase and consumption growth was still higher than income growth. This was because income accelerated sufficiently fast to allow debt servicing and consuming at the same time. However, such a situation is hardly likely because a large positive productivity shock is required in this case. It is more common that savings would likely rise during deleveraging.

3. Monetary Policy Decision

Thailand's economic growth improved further driven by external demand, together with a continued recovery in domestic demand that began to be more broad-based. However, supporting factors to domestic demand were not yet robust. An accommodative monetary policy stance was thus still warranted.

The Committee assessed that Thailand's growth outlook continued to improve driven primarily by a better-than-expected growth in merchandise exports and tourism underpinned by a stronger global economic recovery. The positive effects from the external front were started to be observed in some related sectors with signs of improvements in credit utilization and credit quality of large businesses as well as employment in export-related and tourism businesses. Meanwhile, private consumption was expected to gradually expand given that purchasing power was not yet robust, particularly of low-income households. Private investment in machinery and equipment improved across various businesses, partly attributable to improved economic and investment sentiments. Moreover, public expenditure still remained a growth driver but would likely slow down somewhat due to prior accelerated disbursement in the previous period and some government agencies that were constrained by limited disbursement efficiency.

Nevertheless, while the Thai economy exhibited a stronger growth outlook on the back of both external and domestic factors with greater positive spillovers across various sectors of the economy, the expansion was not yet sufficiently strong. This was reflected in employment of low-income workers that did not yet broadly recover and a decline in credit quality of SMEs. In addition, export growth still faced significant risks from the external front which included uncertainties surrounding U.S. economic and foreign trade policies, concerns on China's financial stability, and heightened geopolitical risks which could undermine private sector confidence and financial conditions. Thus, the Committee viewed that accommodative monetary policy stance was still necessary to support economic growth to continue and strengthen. The Committee would continue to monitor the positive spillovers from increasing economic activities to other sectors of the economy in order to formulate appropriate monetary policy in the period ahead.

Headline Inflation was expected to slowly rise albeit at a slower pace than previously assessed due to supply-side and structural factors. Hence, further monetary policy accommodation would yield a limited impact in boosting inflation.

Headline inflation would trend up slowly in tandem with economic growth; at the same time, last year's high base effects following the drought gradually dissipated. However, the Committee projected that the average annual headline inflation might be lower than the assessment in the previous *Monetary Policy Report*. This was owing to a larger-than-expected decline in fresh food prices as a result of a higher output of vegetable and fruits amid favorable weather conditions. Meanwhile, core inflation which reflected demand-pull inflationary pressures remained stable at a low level.

The Committee viewed that headline inflation that was below the lower bound of the target in recent periods was caused by supply-side factors, particularly lower fresh food prices that were in line with prices of vegetable and fruits on account of higher output amid favorable

weather conditions. These led the food components in core inflation such as prepared food to stabilize at a low level (Chart 2.1 2.2 and 2.3 in Chapter 2). Moreover, low headline inflation was also attributable to structural factors which included the intensified competition following globalization and e-commerce such that both domestic and foreign firms continued to increase sales promotion as well as technological advancements that led to lower costs of production and services. This situation, where headline inflation was lower than target because of structural factors, was also observed in many countries. Against this backdrop, the impact of further monetary policy accommodation in boosting headline inflation would be limited. Nevertheless, the current level of monetary policy accommodation should be maintained to support domestic demand growth which would consequently facilitate a rise in inflation, although this might take some time. Looking ahead, uncertainties surrounding the inflation outlook could stem from the impact of regulations on immigrant workers, which might cause wages to rise as a result of a tightening labor market, and an increase in excise tax. These factors could somewhat put upward pressures on inflation. The Committee would thus continue to closely monitor inflation developments and continue to communicate with public to anchor the public's inflation expectations within the target band going forward.

Financial conditions remained accommodative as reflected in a new loan rate (NLR) which remained stable at a low level after having declined in the previous period. Meanwhile, the baht appreciated against the U.S. dollar in line with most regional currencies.

The Committee assessed overall financial conditions to be accommodative and conducive to economic growth, as reflected in high liquidity in the financial system and the low real policy rate. Business financing increased through both credit and capital markets. With regard to the bond markets, short-term government bond yields were below the policy rate in recent periods, due to the reduction in short-term bond issuances by the Bank of Thailand and the government, and did not indicate future monetary policy stance. Nevertheless, short-term yields later picked up somewhat due to portfolio adjustments by some investors and expectations that the government would resume issuing treasury bills at the beginning of the next fiscal year.

Nevertheless, overall business financing costs through commercial banks were accommodative as reflected in the new loan rate (NLR) that remained stable at a low level after trending down in the previous period (Chart 2.7 in Chapter 2). Meanwhile, credit lines of businesses were not fully utilized especially those of large businesses, reflecting demand for business credits that only increased moderately. In addition, there were still problems mostly regarding access to credits faced by SMEs which necessitated the use of other measures in solving this problem other than monetary policy.

With regard to exchange rates, the baht appreciated against the U.S. dollar relative to the previous quarter in line with most regional currencies due to (1) the weakening of the U.S. dollar from uncertainties pertaining to U.S. politics and stimulus economic policies and weaker-than-expected U.S. economic data outturns, (2) a higher-than-expected Thailand's current account surplus, and (3) capital inflows to Thailand through overseas fundraising by Thai corporates and foreign affiliates as well as portfolio investment in the Thai stock market. Thus, the baht recorded a stronger appreciation relative to regional peers in some periods. Nevertheless, the pace of the baht appreciation slowed down somewhat at the end of the third quarter. The Committee viewed that the baht's movement relative to those of trading partners

was mostly unchanged. As of the end of September 2017, although the baht recorded a net appreciation of 8 percent from the end of 2016, the Real Effective Exchange Rate (REER) only strengthened by 2 percent and was ranked in the middle among regional peers (Chart 2.12 in Chapter 2). Looking ahead, exchange rates might experience high volatilities due to uncertainties on the external front, such as uncertainties pertaining to U.S. economic policies and monetary policy conducts of major advanced economies. The Committee would thus continue to closely monitor developments in the foreign exchange markets.

Financial stability remained sound overall but there remained pockets of risks that might pose vulnerabilities to financial stability under a prolonged low interest rate environment.

The Committee viewed that overall financial stability remained sound and able to provide cushion against volatilities in global and domestic financial markets. Nevertheless, there remained pockets of risks that might pose vulnerabilities to financial stability that warranted monitoring going forward. These included **a rising search-for-yield behavior** in a prolonged low interest rate environment which might result in underpricing of risks**.** The Committee would continue to monitor developments of important risks, such as the continued expansion in foreign investment funds (FIFs) that was concentrated only in some countries, although risks were limited as these funds invested in countries with investment-grade credit ratings, as well as a search-for-yield behavior of saving cooperatives following an expansion in members' deposits. Another risk was the **deterioration of debt serviceability of households and SMEs** stemming from elevated debt levels and changes in structural factors and business models that might weigh on SMEs' adjustments amid a changing environment in business competitions. Meanwhile, the situation regarding the rollover of unrated corporate bonds improved, but there were still risks that warranted monitoring including debt serviceability of bonds that would mature in the period ahead, especially those of issuers with fragile financial positions. Risk could also stem from the situation regarding an **oversupply of condominiums in Bangkok and its vicinity**. Although overall risks were stable, a situation regarding an increasing number of unsold units of condominiums in certain areas and price ranges and a longer "time to go" must be monitored. The Committee underlined the importance of developments in financial market infrastructure and collaboration among regulatory authorities in order to enable better information access for risk assessment and prevent any exploitation of the regulatory gap which might pose systematic risks. The Committee would stand ready to implement appropriate macroprudential measures in a timely manner.

The Committee concurred that monetary policy should remain accommodative to support the continuation of economic growth, which would in part facilitate the increase in the inflationary pressures in the period ahead.

The Committee voted unanimously to keep the policy interest rate at 1.5 percent at the meetings on August 16 and September 27. Thailand's economic growth outlook continued to improve on the back of both domestic and external factors with greater positive spillovers across various sectors of the economy. Inflationary pressures, while remaining low due to supply-side and structural factors, were expected to gradually rise in tandem with an economic recovery in the period ahead. However, domestic demand expansion was not yet strong and the economy still faced external risks that might hamper export and tourism growth. Nevertheless, financial conditions that remained accommodative for extended period might exacerbate risks to financial stability. Thus, certain risks must be monitored, including impacts of the prolonged low interest

rate environment on national savings as well as a buildup of financial system vulnerabilities from the search-for-yield behavior which might lead to a widespread underpricing of risks. Therefore, the Committee viewed that the current degree of monetary policy accommodation should be maintained so that the economic growth could continue and strengthen, which would facilitate the return of headline inflation to target in the period ahead. The Committee would stand ready to utilize an appropriate mix of available tools to support the economic growth while also preserving price and financial stability.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2015	2016	2016			2017	
			Q2	Q3	Q4	Q1	Q2
GDP growth	2.9	3.2	3.6	3.2	3.0	3.3	3.7
Production							
Agriculture	-5.7	0.6	-0.4	0.9	3.0	5.7	15.8
Non-agriculture	3.9	3.5	3.9	3.2	3.2	3.1	2.7
Manufacturing	1.5	1.4	2.2	1.6	2.2	1.3	1.0
Construction	17.0	8.3	9.9	5.2	6.1	2.8	-6.2
Wholesales and retail trade	3.9	5.0	4.7	5.2	5.6	5.9	6.0
Hotels and restaurants	14.6	10.3	10.8	13.5	4.9	5.3	7.5
Transport, storage, and communication	5.1	5.6	4.2	6.5	5.2	5.4	8.6
Financial intermediation	8.8	6.1	6.2	5.8	6.7	4.6	5.1
Real estate, renting, and business activities	1.9	1.8	2.1	1.1	1.9	4.0	4.1
Expenditure							
Domestic demand	2.9	2.7	3.5	0.9	2.2	2.3	2.3
Private consumption	2.2	3.1	4.0	3.0	2.5	3.2	3.0
Private investment	-2.2	0.4	0.3	-0.8	-0.4	-1.1	3.2
Government consumption	3.0	1.7	2.4	-5.2	1.8	0.3	2.7
Public investment	29.3	9.9	12.8	5.8	8.6	9.7	-7.0
Imports of goods and services	0.0	-1.4	-2.5	-1.1	3.4	6.1	8.2
imports of goods	0.2	-2.1	-2.8	-1.5	3.6	7.3	9.1
imports of services	-1.0	1.7	-1.3	0.5	2.0	1.1	4.0
Exports of goods and services	0.7	2.1	1.0	1.4	1.1	2.7	6.0
exports of goods	-3.4	0.0	-1.9	-0.4	1.4	2.6	5.2
exports of services	17.1	9.3	11.6	7.7	0.4	3.2	8.8
Trade balance (billion, U.S. dollars)	26.8	36.5	8.5	9.2	7.1	8.8	6.4
Current account (billion, U.S. dollars)	32.1	48.2	9.0	11.7	10.8	15.0	7.4
Financial account (billion, U.S. dollars)	-16.8	-21.0	-1.3	-7.8	-12.1	-7.0	-4.1
International reserves (billion, U.S. dollars)	156.5	171.9	178.7	180.5	171.9	180.9	185.6
Unemployment rate (%)	1.0	1.0	1.1	0.9	1.0	1.2	1.2
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.0	1.0	1.1	1.1	1.1

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2015	2016	2016		2017			
			Q3	Q4	Q1	Q2	Jul	Aug
1. Financial market sector								
Bond market								
Bond spread (10 years - 2 years)	1.1	0.6	0.6	0.8	1.1	1.1	1.0	1.0
Equity market								
SET index (end of period)	1,288.0	1,542.9	1,483.2	1,542.9	1,575.1	1,574.7	1,576.1	1,616.2
Actual volatility of SET index[1/]	13.9	15.2	13.6	15.2	7.0	4.8	3.7	7.3
Price to Earnings ratio (P/E ratio) (times)	22.6	18.6	21.3	18.6	17.4	16.3	16.4	17.2
Exchange rate market								
Actual volatility of Thai baht (%annualized)[2/]	5.1	4.4	3.8	5.0	3.5	3.9	3.1	3.2
Nominal Effective Exchange Rate (NEER)	108.5	106.2	105.8	107.0	108.7	109.8	110.9	111.4
Real Effective Exchange Rate (REER)	104.4	100.6	100.4	101.1	102.1	102.7	103.6	n.a.
2. Financial institution sector[3/]								
Minimum Lending Rate (MLR)[4/]	6.50	6.26	6.26	6.26	6.26	6.20	6.20	6.20
12-month fixed deposit rate[4/]	1.40	1.38	1.38	1.38	1.38	1.38	1.38	1.38
Capital adequacy								
Capital funds / Risk-weighted asset (%)	17.4	18.0	18.5	18.0	17.8	17.9	n.a.	n.a.
Earning and profitability								
Net profit (billion, Thai baht)	192.3	199.2	49.8	47.4	51.2	49.0	n.a.	n.a.
Return on assets (ROA) (times)	0.9	1.1	1.2	1.2	1.2	1.2	n.a.	n.a.
Liquidity								
Loan to Deposit and B/E (%)	97.0	96.3	96.9	96.3	95.7	96.5	n.a.	n.a.
3. Household sector								
Household debt to GDP (%)	81.2	79.9	80.0	79.9	78.7	78.4	n.a.	n.a.
Financial assets to debt (times)	2.6	2.8	2.57	2.8	2.7	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Consumer loans	2.6	2.7	2.7	2.7	2.8	2.7	n.a.	n.a.
Housing loans	2.4	2.9	2.8	2.9	3.2	3.1	n.a.	n.a.
Auto leasing	2.3	1.8	1.8	1.8	1.6	1.7	n.a.	n.a.
Credit cards	4.0	3.7	5.1	3.7	4.1	3.4	n.a.	n.a.
Other personal loans	2.7	2.9	2.9	2.9	2.9	2.6	n.a.	n.a.
4. Non-financial corporate sector[5/]								
Operating profit margin (OPM) (%)	7.4	8.3	8.2	7.7	8.5	7.5	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	5.7	6.6	5.8	6.8	6.2	6.3	n.a.	n.a.
Current ratio (times)	1.7	1.6	1.7	1.6	1.7	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)								
Large businesses	1.6	1.5	1.9	1.4	1.6	1.8	n.a.	n.a.
SMEs	3.5	4.3	4.0	4.3	4.5	4.4	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2015	2016	2016		2017			
			Q3	Q4	Q1	Q2	Jul	Aug
5. Real estate sector								
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)								
Total	59,667	61,452	14,149	16205	12244	15,086	4,733	5,724
Single-detached and semi-detached houses	13,152	13,409	3,216	3179	2802	3,544	1,073	1,236
Townhouses and commercial buildings	19,210	20,187	4,718	4967	4315	4,947	1,638	1,914
Condominiums	27,305	27,856	6,215	8059	5127	6,595	2,022	2,574
Number of new housing units launched for sale (Bangkok and Vicinity) (units)								
Total	107,988	110,575	31,017	31,452	25,304	25,529	6,375	12,302
Single-detached and semi-detached houses	17,637	19,433	6,328	4,973	2,054	2,413	881	1,423
Townhouses and commercial buildings	27,518	32,792	10,638	7,861	10,413	7,102	1,254	4,763
Condominiums	62,833	58,350	14,051	18,618	12,837	16,014	4,240	6,116
Housing price index (2009 = 100)								
Single-detached houses (including land)	129.3	128.8	131.3	128.8	128.6	129.6	130.6	130.9
Townhouses (including land)	137.5	135.6	136.8	135.6	138.3	140.0	141.0	141.4
Condominiums	160.9	173.6	169.2	173.6	169.8	168.8	168.6	168.4
Land	168.8	171.3	170.2	171.3	171.3	164.2	167.0	169.0
6. Fiscal sector								
Public debt to GDP (%)	44.4	42.2	42.8	42.2	42.2	41.8	41.8	41.9
7. External sector								
Current account balance to GDP (%)[6/]	8.1	11.7	10.9	9.7	14.1	7.7	n.a.	n.a.
External debt to GDP (%)[7/]	32.0	32.7	34.8	32.7	33.5	34.3	n.a.	n.a.
External debt (billion, U.S. dollars)	131.1	132.2	140.6	132.2	136.2	140.5	142.9	146.5
Short-term (%)	40.1	41.2	41.0	41.2	40.5	39.5	39.5	39.7
Long-term (%)	59.9	58.8	59.0	58.8	59.5	60.5	60.5	60.3
International reserves / Short-term external debt (times)	3.0	3.2	3.1	3.2	3.3	3.3	3.4	3.4

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2017		2018				2019	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
10-12	0	0	0	0	0	0	1	1
8-10	0	0	1	2	2	3	4	5
6-8	2	9	10	10	12	12	13	13
4-6	37	39	32	27	26	25	24	23
2-4	51	38	35	31	29	28	26	25
0-2	9	12	18	20	19	20	19	18
(-2)-0	0	1	4	8	8	9	10	10
< (-2)	0	0	1	2	3	4	4	5

Table: Probability distribution of core inflation forecast

Percent	2017		2018				2019	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
3.5-4.0	0	0	0	0	0	0	0	0
3.0-3.5	0	0	0	0	0	0	1	1
2.5-3.0	0	0	0	2	2	2	3	4
2.0-2.5	0	0	2	6	7	6	7	9
1.5-2.0	0	5	10	15	15	13	14	15
1.0-1.5	5	23	24	23	22	20	20	20
0.5-1.0	44	41	31	24	23	23	21	20
0.0-0.5	45	26	21	17	17	19	17	15
(-1)-0.0	5	6	8	8	9	11	10	9
(-2)-(-1)	0	0	2	3	3	5	4	4
< -2	0	0	0	1	1	1	1	1

Table: Probability distribution of headline inflation forecast

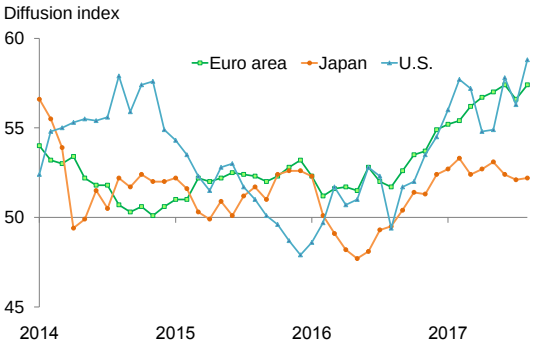
Percent	2017		2018				2019	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 7	0	0	0	0	0	0	0	1
6-7	0	0	0	0	1	0	0	1
5-6	0	0	0	2	2	2	2	2
4-5	0	0	1	5	5	5	4	5
3-4	0	1	4	11	11	10	9	10
2-3	1	7	12	18	18	16	16	16
1-2	17	24	23	22	21	20	20	20
0-1	50	36	27	19	19	20	19	19
(-1)-(0)	28	24	20	13	13	14	15	14
(-2)-(-1)	3	7	9	6	6	8	9	8
(-3)-(-2)	0	1	3	2	2	3	4	4
(-4)-(-3)	0	0	0	1	1	1	1	1
< (-4)	0	0	0	0	0	0	0	0

4.2 Chartpack

The Global Economy

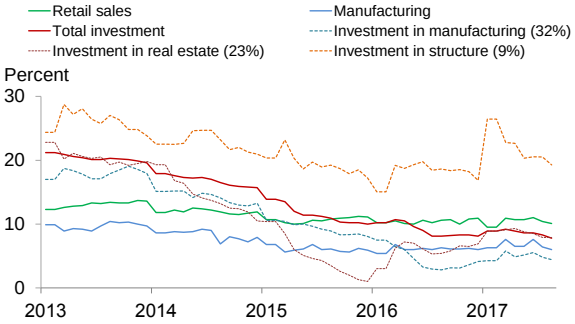
Advanced economies continued to expand further driven by expansion in consumption and stronger labor markets. China continued expanding on the back of investment in the first half of the year, but was projected to slightly slow down as economic structural reforms began to bear results. Meanwhile, Asian exports steadily improved in line with global demand and the technology cycle of electronics.

Manufacturing Purchasing Manager Index



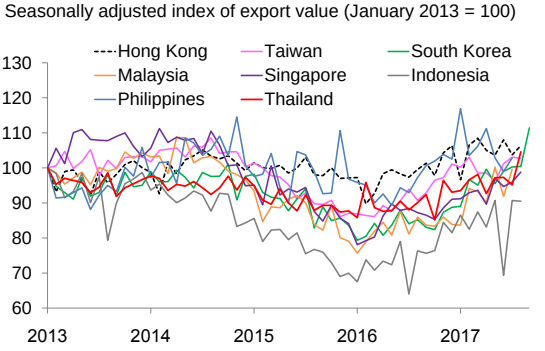
Source: Bloomberg

China's economic indicators (Change from the same period last year)



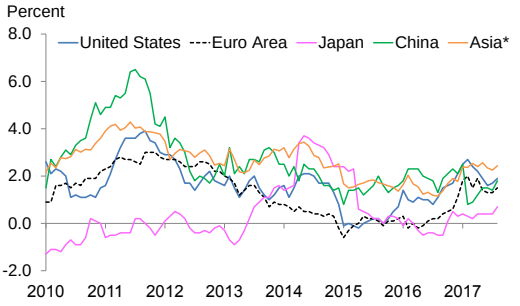
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

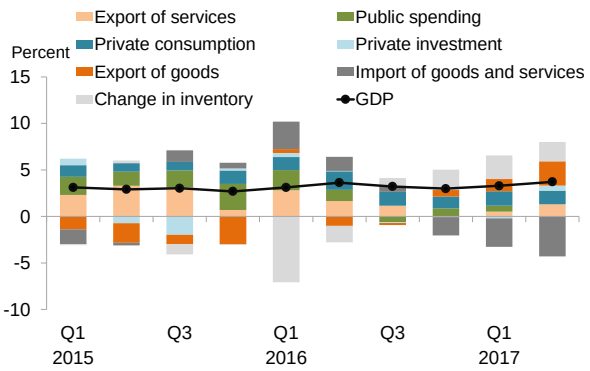


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

The Thai Economy

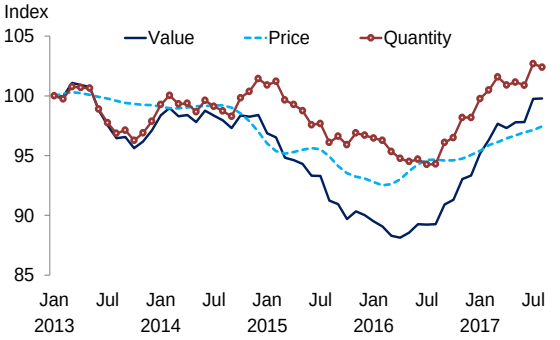
Thailand's economic growth gained further traction in the second quarter of 2017, underpinned by exports and tourism with signs of greater positive spillovers to domestic consumption and investment. Meanwhile, the number of foreign tourists continued to steadily increase. Public spending remained an important growth driver but would likely slow down due to prior accelerated disbursement in the previous period and some government agencies that were constrained by disbursement efficiency.

Contribution to Thailand's GDP growth[1/]



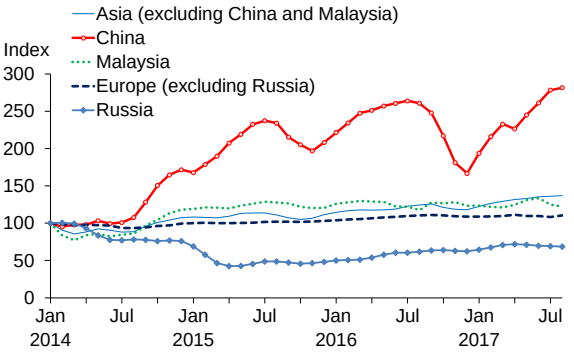
Note: [1/] Calculated by Chain Volume Measure method (CVM)

Thai exports (excluding gold): value, price and quantity
(3-month moving average, seasonally adjusted; January 2013 = 100)



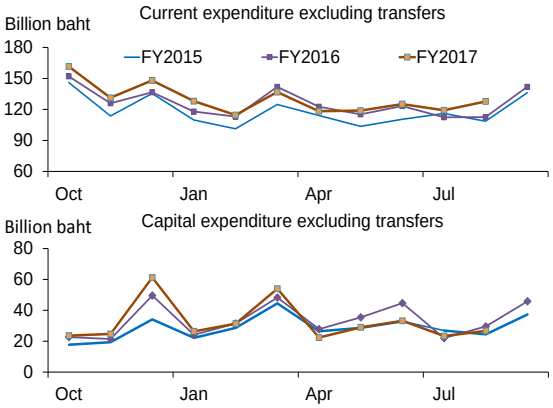
Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Index of foreign tourists classified by nationality
(3-month moving average, sessionally adjusted; January 2014 = 100)



Source: Department of Tourism

Public spending by central government

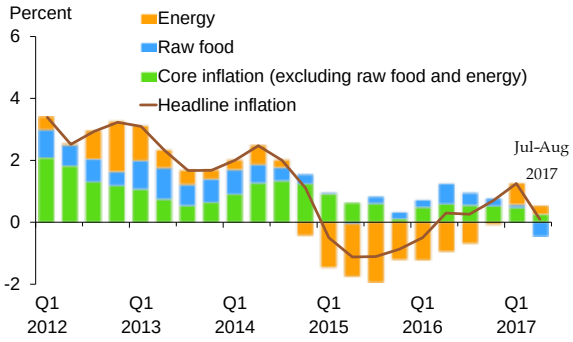


Source: Bureau of Budget, Fiscal Policy Office

Inflation

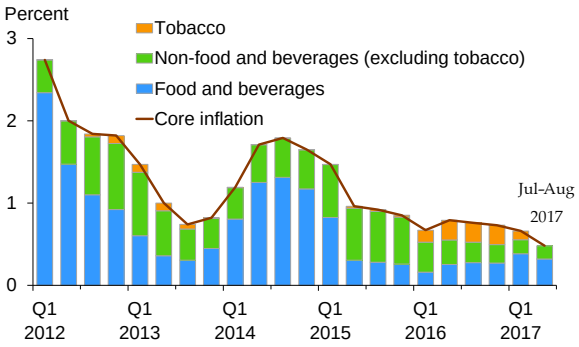
Headline inflation slightly edged up mainly due to energy prices. Core inflation remained stable at a low level in line with a gradual expansion of domestic demand. Meanwhile, five-year-ahead inflation expectations by professional forecasters dropped slightly, but remained close to the midpoint of the inflation target.

Contribution to headline inflation



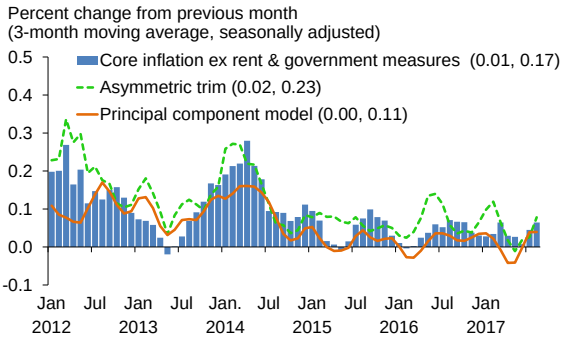
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

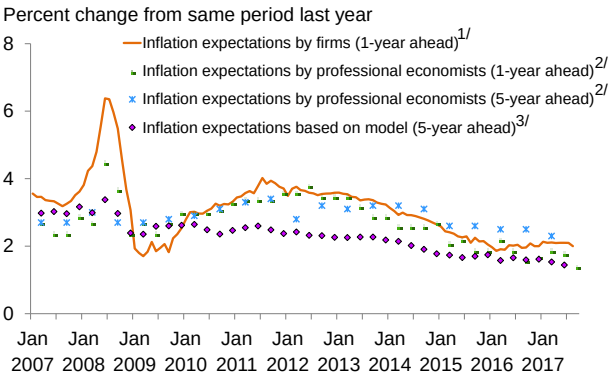
Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of May 2017, while the second value is 2004 - 2014 average; Asymmetric trim excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; Principal component model calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

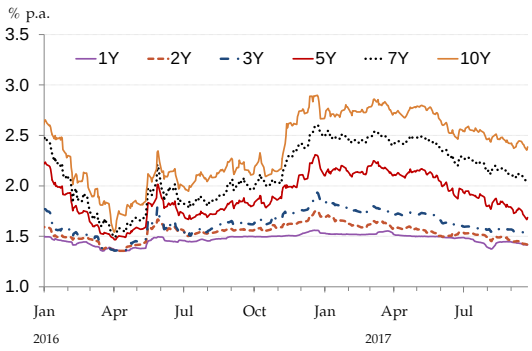


Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial conditions

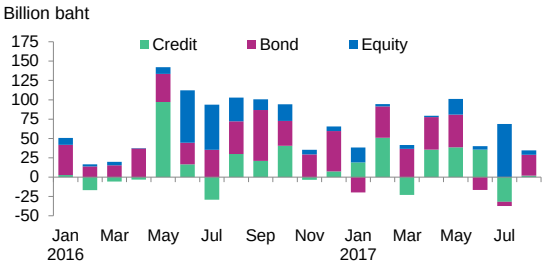
Medium- to long-term government bond yields declined due to external factors. Total corporate financing continued to increase mainly through equity and bond markets. The Thai baht appreciated against the U.S. dollar, while the baht's movement relative to those of trading partners was largely unchanged.

Thai government bond yields



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

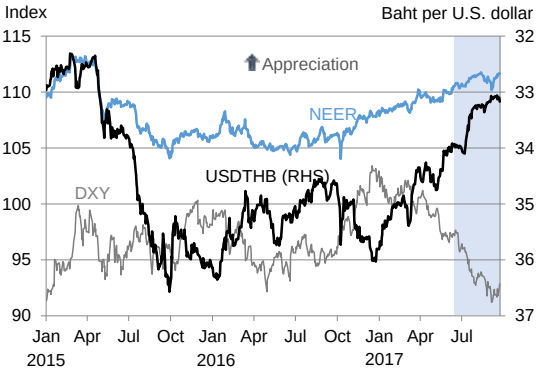
Total corporate financing by instruments*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bond excluding commercial banks, and newly issued equities.

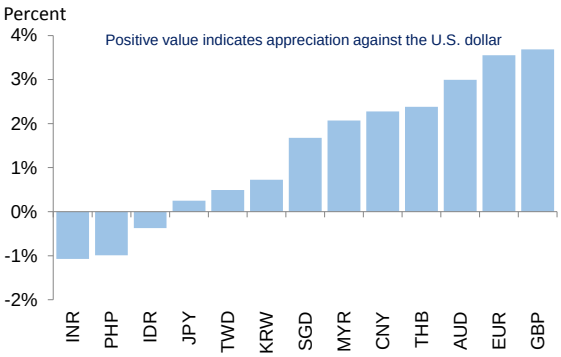
Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 26 September 2017)

Currency movements vis-a-vis the U.S. dollar (26 Sep 2017 compared to 30 June 17)

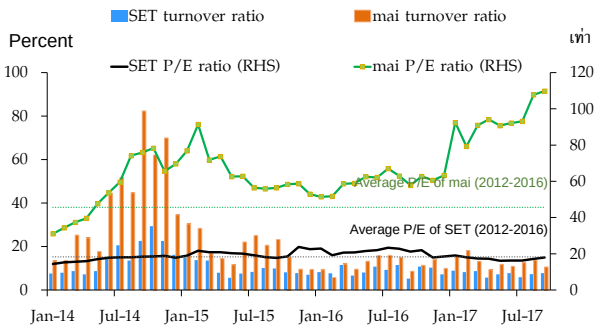


Source: Bank of Thailand and Reuters (data as of 26 Sep 2017)

Stability: financial markets

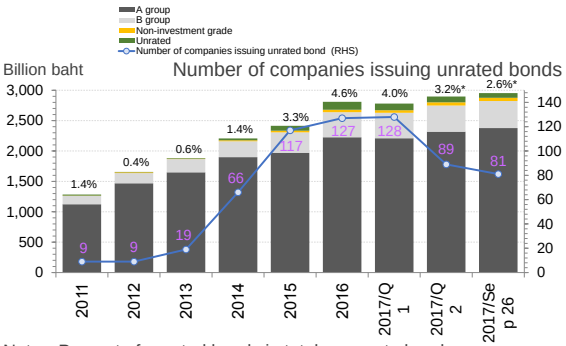
The price-to-equity (P/E) ratio of the Stock Exchange of Thailand stayed close to the historical average. The P/E ratio of the Market for Alternative Investment (mai) registered a new record high due to losses in agricultural and financial sectors in the second quarter. Meanwhile, issuances of unrated bonds steadily declined since the end of 2016 after some defaults. Unrated bonds still constituted a small portion in the total corporate bonds outstanding.

Current Price to earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of 26 Sep, 2017)

Corporate bonds outstanding

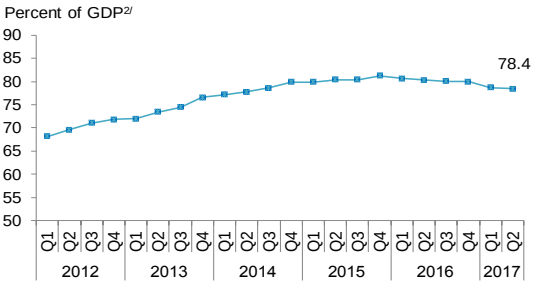


Note: * Percent of unrated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: the household sector

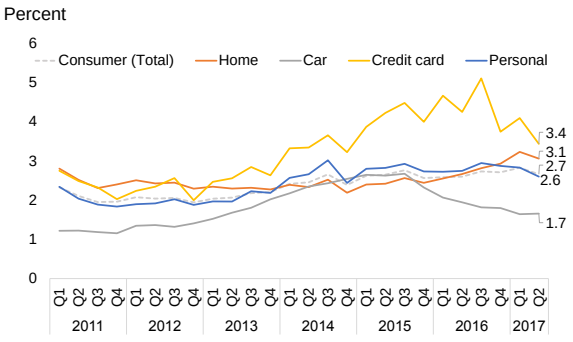
The ratio of household debt to GDP continued to trend down. Deleveraging was observed in some high-income groups and more evidently for automobile loans. However, deteriorating household debt serviceability warranted close monitoring.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
Source: Bank of Thailand

Share of non-performing loans (NPLs) in consumer loans, classified by loan type

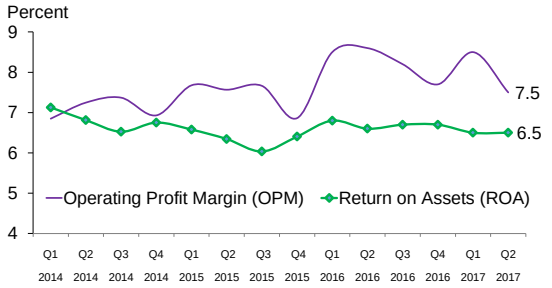


Source: Bank of Thailand

Stability: the corporate sector

Overall stability of the corporate sector remained sound. However, several corporate sector indicators showed that overall economic growth was not broad-based, as reflected by deterioration in income prospects, profitability, and debt serviceability of small businesses.

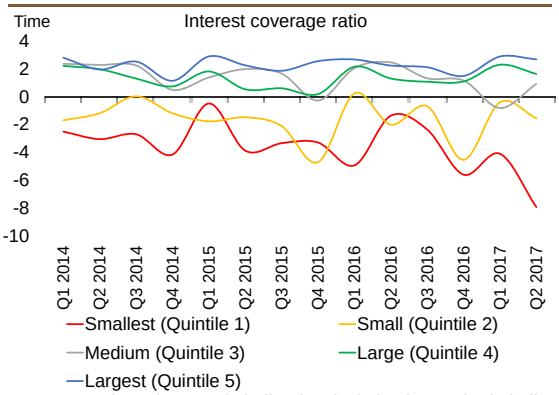
Operating Profit Margin (OPM) and Return on Assets (ROA)*



Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.

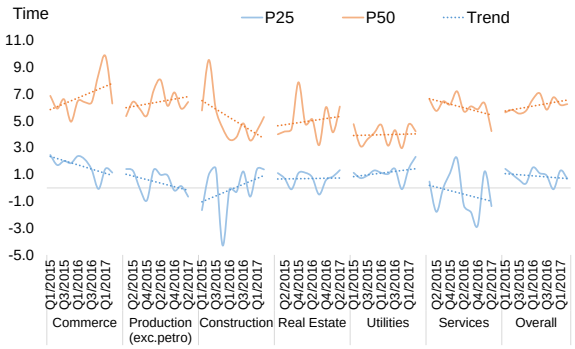
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



Source: Stock Exchange of Thailand, calculation by Bank of Thailand

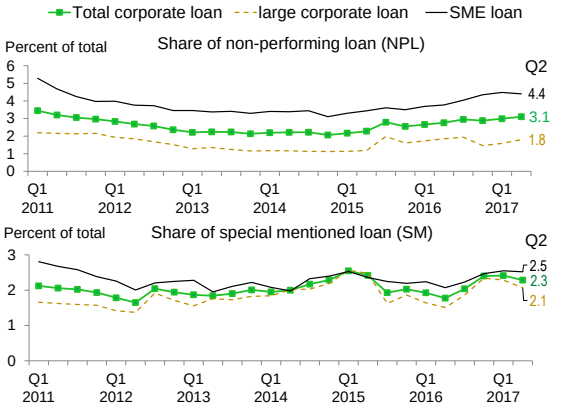
Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals

Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

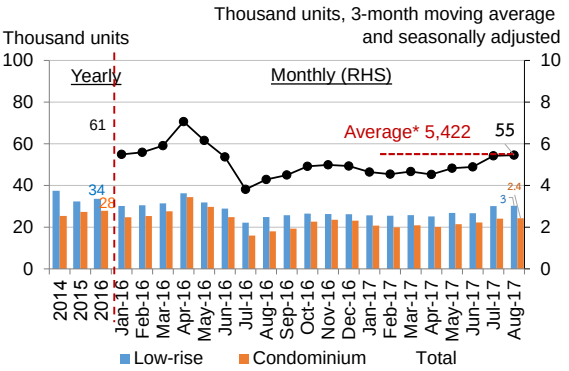


Source: Bank of Thailand

Stability: the real estate sector

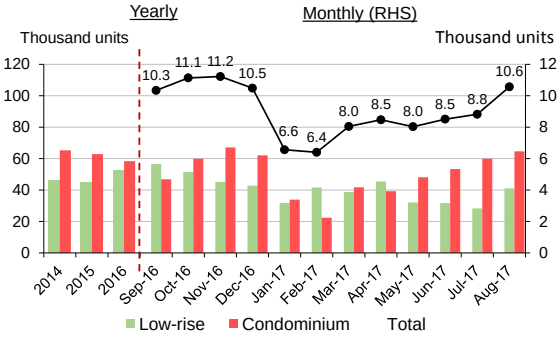
The real estate sector overall improved. Demand for condominiums and low-rise residences in Bangkok and its vicinity increased from the previous quarter. On the supply side, developers continued to launch new residential projects, especially condominiums. Residential property prices remained broadly stable.

Residential units in Bangkok and its vicinity with approved mortgages by commercial banks



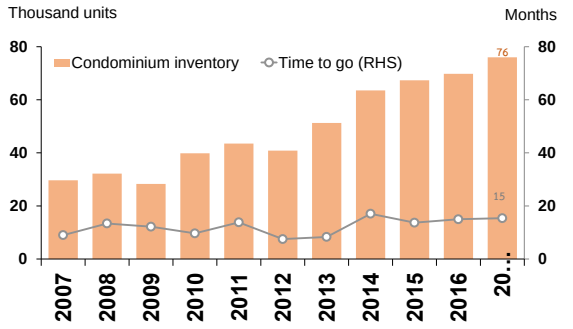
Note: *Average during 2010-2016, excluding periods with government's stimulus measures (November 2015-April 2016)
Source: Bank of Thailand

New residential projects launched in Bangkok and its vicinity



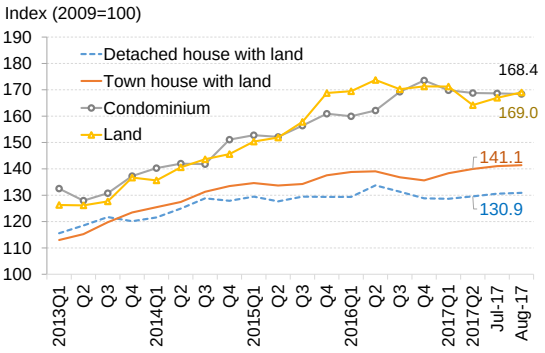
Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

Condominium inventory in Bangkok and its vicinity and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold, using the average sales rate of the past 12 months.
Source: AREA and calculations by Bank of Thailand

Real estate price indices

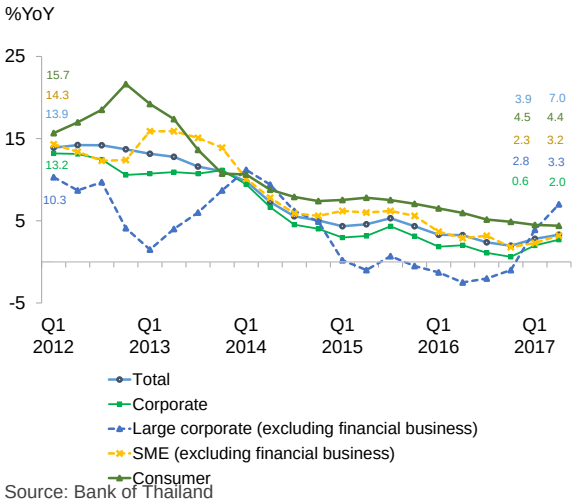


Source: Bank of Thailand

Stability: financial institutions

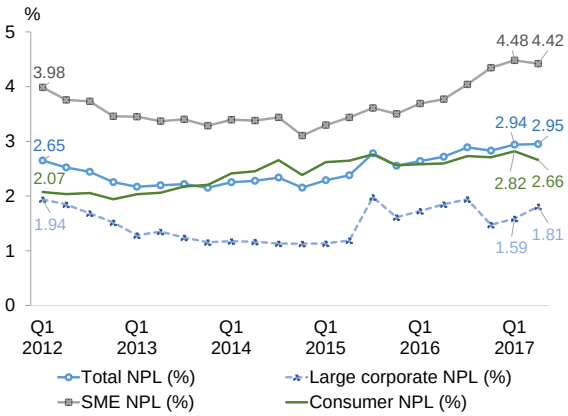
Credit growth rose in the second quarter primarily due to loans to large corporates. Meanwhile, loans to households remained stable. The quality of corporate loan deteriorated somewhat due to certain large businesses, while the quality of household loans improved. Nonetheless, the financial system remained sound with high levels of provisions and capital buffers.

Credit growth in the commercial bank system



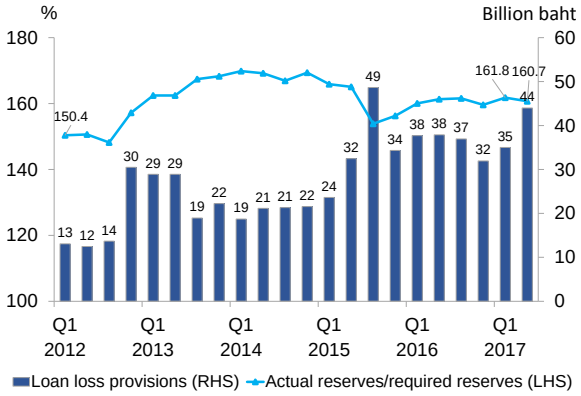
Source: Bank of Thailand

Non-performing loan (NPL)



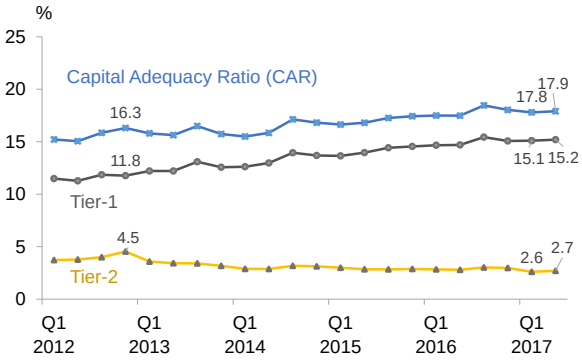
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

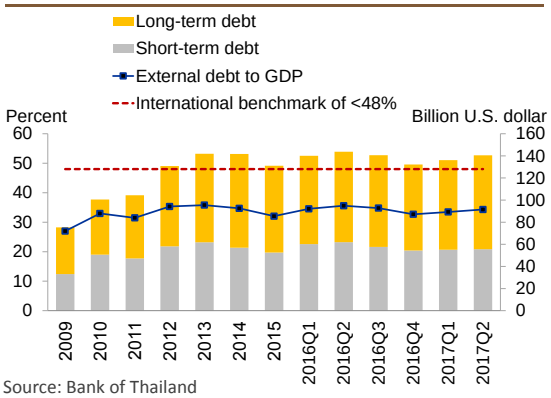


Source: Bank of Thailand

Stability: external position

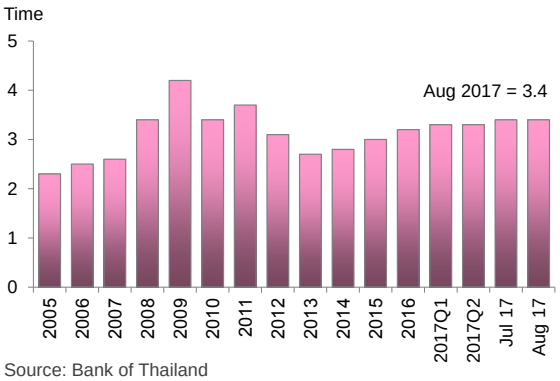
Thailand's external stability remained strong due to a lower level of external debt than an international benchmark and international reserves were at a high level relative to short-term debt.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

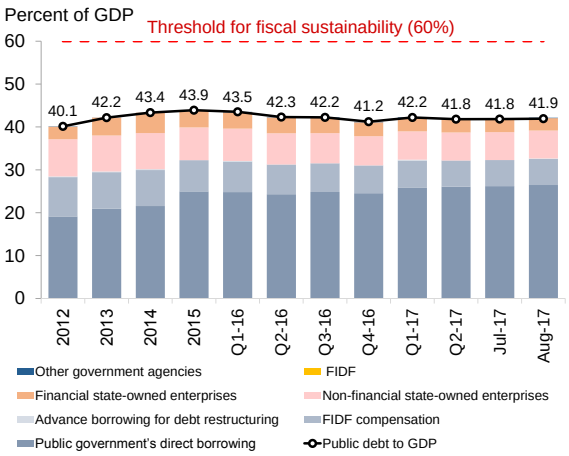


Source: Bank of Thailand

Stability: the fiscal sector

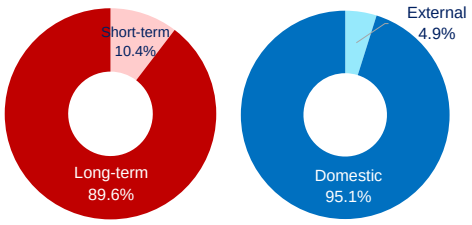
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of August 2017



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office

Pursuing Sustainable
Economic Well-Being

Bank of Thailand
273 Samsen Road, Watsamphraya, Phra Nakhon District, Bangkok 10200



Bank of Thailand Channel
Hotline1213 Channel



www.bot.or.th



@bankofthailand



bankofthailand
Hotline1213